ChemicalQDevice | Independent Research Paper | Draft 1.0

From Independent Scientist to Novel Performer

A Small-Business Operating, Milestone, and Capitalization Plan for a Phase 1 LLM-Advised Robotic Whipple (ChemicalQDevice)

Draft 1.0 | **Repository v4.5.0** | **20 figures** | **12 milestones**

SBIR Phase I	SBIR Phase II	Five-year programme
$306,000	**$1,300,000**	**$3,500,000**
9 months, 5 milestones	24 months, 7 milestones	60 months, direct

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Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

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The paper (v1.0) is at `https://doi.org/10.5281/zenodo.xxxxxxxx` and the repository (v4.5.0) is at `https://github.com/kevinkawchak/physical-ai-oncology-trials`. The paper files are deposited in funding/capitalization-plan.

Keywords Small business innovation research; Novel performers; Capitalization plan; Milestone schedule; Indirect cost; 21 CFR part 54; Pancreatic ductal adenocarcinoma; Robotic pancreaticoduodenectomy; Daraxonrasib; Physical AI

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Abstract

In August 2026 the author filed ten funding applications for a Phase 1 trial of an LLM-advised robotic pancreaticoduodenectomy with perioperative daraxonrasib, written throughout as an independent scientist [1]. Nine of the ten address mechanisms that fund a person or an institution. One, application 05, addresses the only mechanism in the set built for a company, and it was the shortest of the ten. This paper converts the author from independent scientist to small-business operator and rewrites that application as the document it should have been.

The clause the conversion turns on is neither the report's individual-scientist language nor its novel-performers chapter. It is the Small Business Innovation Research clause of *Science: A New Golden Age*, which appears in three separate chapters and is the only clause in the report written for a firm [2]. Three numbers govern what follows: $306,000 for a nine-month Phase I, $1,300,000 for a twenty-four-month Phase II, and the $3,500,000 five-year direct programme the two are measured against [3]. Two more are derived: $1,396,000 of direct work sits inside the $1,606,000 award, and $2,104,000 sits outside it.

What is offered is a costed twelve-milestone schedule. Every row carries an evidence artifact that exists as a file, a cost that sums into its phase, a stop condition stated before the work begins, and the funder mechanism that pays.

Executive Summary

The clause. *Science: A New Golden Age* names SBIR three times: in the Chapter I summary of recommendations, in Chapter III on deploying SBIR and STTR strategically to couple federally-seeded companies with the scientific enterprise, and in Chapter IV on SBIR opening doors for technician-founded ventures. Its novel-performers chapter describes work at tens of millions of dollars with teams of ten to a hundred people over half a decade. This company proposes $1,606,000, 2.6 full-time equivalents, and 33 months.

The entity. ChemicalQDevice holds thirteen deposited assets outright and none under encumbrance: two protocols, an IND package, a simulation stack, a verification suite, and a public repository. It licenses three things, none exclusively. It has contracted for nothing. Seven instruments are absent, and not one can be obtained by the company alone.

The two prices. The identical four-layer budget costs $3,500,000 of direct work over sixty months and $1,396,000 of direct work over thirty-three. The award that carries the second is $1,606,000, a 15.0 percent overhead and fee load against an SBIR allowance of 40 percent that the company does not claim. Routed through a university at a 57 percent facilities and administrative rate, the same direct work would cost a funder $2,137,000.

The bridge. Federal money of $1,606,000 sits below a firewall; contributed drug, theatre and pathology sit unpriced in the middle; and $5,900,000 of private capital sits above, raised only after a milestone closes. That is 3.67 to one on cash alone, against the annexed FY 2028 target of at least three to one [4].

The milestones. Twelve rows, five in Phase I summing to $306,000 and seven in Phase II summing to $1,300,000. Each is closed by three concurrent evidence branches that must join before an audit can compare them.

What the conversion costs. The individual-scientist framing that made all ten prior applications distinctive is abandoned. Capitalization detail the author might prefer private is published permanently under a DOI. Three of the ten prior recipients will read a company plan as off-mechanism. Those four concessions are stated in §9 and are not softened anywhere.

How to Read This Document

The document is written so that four different readers can each open it at a different page and find the thing they came for within two pages.

Reader	Open at	Then	The question this reader is actually asking
SBIR program director	§3, Table 11	§5, Tables 14 and 15	Does the award buy a defensible gate, and does the budget sum?
ARPA-H programme manager	§5, Figure 13	§1, Figure 1	Is there a milestone I can hold someone to, and a stop condition?
Private investor	§4, Table 13	§2, Table 6	What do I own, when may I own it, and what is already built?
UC San Diego intake	§8, Figure 19	§7, Figure 18	What is being asked of this institution, and by when?

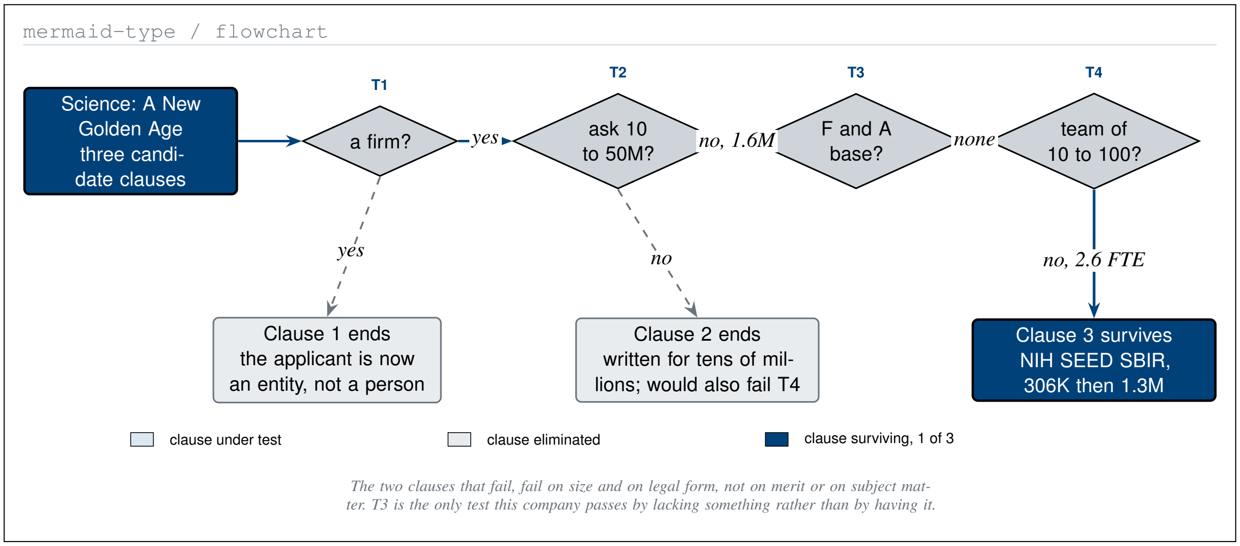
Four readers and the section each should open first. No reader is asked to read the document front to back, and every entry point is a table or a figure rather than a paragraph.

Fig	§	Platform	Source directory	The question it answers
1	1	Mermaid	`mermaid/`	Which clause does a company with no indirect-cost base qualify under?
2	1	D2	`d2/`	What kind of institution is a 2.6 FTE firm, on the report's own axes?
3	1	Graphviz	`graphviz/`	What does the same direct work cost under three overhead regimes?
4	2	Diagrams	`diagrams-python/`	What is the shape of what the company holds today?
5	2	D2	`d2/`	What is owned, licensed, contracted, and absent, with terms?
6	2	PlantUML	`plantuml/`	What may the sponsor do alone, and what needs an institution?
7	3	Mermaid	`mermaid/`	What must hold at month nine for Phase II to begin?
8	3	D2	`d2/`	What does the same programme cost under two mechanisms?
9	3	Graphviz	`graphviz/`	Which work packages does the money reach, and which not?
10	4	PlantUML	`plantuml/`	Which capital positions may be held while a participant is on study?
11	4	D2	`d2/`	Where does every dollar come from, and what separates the tiers?
12	4	Mermaid	`mermaid/`	In what order is a private round executed during enrollment?
13	5	Mermaid	`mermaid/`	When does each milestone open, close, and produce its artifact?
14	5	Graphviz	`graphviz/`	What has to fail, and in what combination, to stop the programme?
15	5	PlantUML	`plantuml/`	What do three parties do at once to close one milestone?
16	6	D2	`d2/`	Which published quantities can a funder check before committing?
17	6	Graphviz	`graphviz/`	Where does each published number land in a filing?
18	7	Diagrams	`diagrams-python/`	Who is employed, who is contracted, who is contributed?
19	8	Mermaid	`mermaid/`	What did the July and August 2026 contacts actually unlock?
20	10	Diagrams	`diagrams-python/`	What survives if the programme stops, and who holds it?

The twenty figures, their section, their platform, the directory that carries each specification, and the one question each answers. Five mermaid, three plantuml, five d2, three diagrams, four graphviz, by purpose not quota.

Tab	§	Contents	The question it answers
1	0	Four readers, four entry points	Which section should this reader open first?
2	0	Figure index, twenty rows	Where is each figure and what does it answer?
3	0	Table index, twenty-one rows	Where is each table and what does it answer?
4	1	Three clauses against four eligibility tests	Which clause survives the filter, and why?
5	1	Three overhead regimes on one direct base	What is the premium for routing through an institution?
6	2	The owned register, thirteen rows with DOIs	What exists today and where is it deposited?
7	2	The absent register, seven rows with signatories	What is missing and who alone can supply it?
8	3	Both awards as direct, indirect and fee	What is an award, and how much of it is work?
9	3	Phase I line items to $306,000	What does nine months of Phase I actually buy?
10	3	Phase II line items to $1,300,000	What does twenty-four months of Phase II buy?
11	3	The four-layer budget under two prices	What does the SBIR route reach, and what does it not?
12	4	The §54.2 and §54.4 disclosure triggers	What must be disclosed, at what threshold, on which form?
13	4	The capitalization table across five stages	Who owns what, and does the SBIR test still pass?
14	5	The twelve milestones, months and cost	What is the milestone, when, and what does it cost?
15	5	Artifact, stop condition, funder mechanism	What proves it, what halts it, and who pays?
16	6	Six checkable quantities with limitations	What can a reviewer verify against a published source?
17	6	Cost and schedule against industry benchmarks	What did the prior work cost, and what does it cost elsewhere?
18	7	The 2.6 FTE by employment status	Who is on the payroll and who is not?
19	9	Ten risks, likelihood, and the mitigation held	What could go wrong and what is already in place?
20	10	The eight build stages and their commits	How was this document produced, and in what order?
21	11	Abbreviations, two column pairs	What does this term mean in this paper?

The twenty-one tables, their section, their contents, and the one question each answers. Every table is set to the full body measure with ragged-right fixed columns, so no cell shows a stretched interword gap.



Three clauses of one report, four eligibility tests, and the only clause a San Diego firm with no indirect-cost base survives. The two that fail, fail on size, not on merit or on subject matter.

1 The Novel-Performer Case

1.1 Three Clauses, One Applicant

Science: A New Golden Age contains three clauses under which a Phase 1 pancreatic cancer trial could plausibly be funded, and they are not interchangeable [2].

The first is the individual-scientist language of Chapter I, which asks the roughly $200 billion annual federal research and development portfolio to stop deferring to "the same incumbents that consume the funding". It is the clause the report supports with the person-based funding comparison and with the productivity slowdown literature [5, 6], and the ten prior applications were written under it [1].

The second is the chapter section headed NOVEL PERFORMERS, which describes "a vast middle ground of mid-scale science", problems "requiring tens of millions of dollars, coordinated teams of ten to a hundred people, and timelines of half a decade".

The third is the Small Business Innovation Research clause, and it is the only one of the three that appears in more than one chapter. The Chapter I summary of recommendations asks agencies to "focus Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs to build strategic capabilities". Chapter III states that "SBIR and STTR programs can be deployed strategically to advance new scientific and technological capabilities, coupling federally-seeded companies with the scientific enterprise". Chapter IV states that "Programs like SBIR open doors for technician-founded ventures, directing resources toward ideas that do not always start with a dissertation".

Test	Question	Basis in the report	ChemicalQDevice, August 2026
T1	Is the applicant a firm rather than a person?	Chapter IV, SBIR opens doors for technician-founded ventures	Yes, a California corporation
T2	Is the ask ten to fifty million?	Chapter II, novel performers is written for tens of millions	No, $1,606,000 over 33 months
T3	Is there an indirect-cost base to recover?	Chapter II, deference to the incumbents that consume the funding	No negotiated rate agreement exists
T4	Is the team ten to a hundred people?	Chapter II, coordinated teams of ten to a hundred	No, 2.6 FTE at full Phase II staffing

The four tests, the sentence in the report that each is drawn from, and this company's answer. Clause 1 fails T1, clause 2 fails T2 and T4, and clause 3 is the only one that survives all four.

1.2 Why the Pioneer Language Does Not Fit an Entity

What changed between the ten applications and this one is the applicant. A person-based mechanism funds a named scientist and follows that person's judgement for five to seven years. ChemicalQDevice is a California corporation with one officer, and a corporation is not a subject that mechanism has a way to fund. Nothing about the science changed; the legal form of the applicant did.

That is a real loss and it is worth naming here rather than in a footnote. The individual-scientist framing is what made all ten prior applications distinctive, and it is the framing the report itself rewards most explicitly. Abandoning it is the first of four costs the conversion carries, and all four are set out in §9 without mitigation.

1.3 Why the Novel-Performers Chapter Does Not Fit Either

The novel-performers chapter is precise about scale, and the precision is what excludes this company. It describes problems requiring tens of millions of dollars, teams of ten to a hundred people, and timelines of half a decade, and it names focused research organizations and the NSF TIP X-Labs as the vehicles built for them. Set against that: $1,606,000, 2.6 full-time equivalents, and 33 months.

The chapter describes a real gap in the federal portfolio. This company does not fill it, and claiming otherwise would be exactly the kind of overreach the report is written against. The same holds for the two executive orders that sit behind the report: the Genesis Mission is a national compute and robotics programme and the gold-standard science order binds all federally funded research, and neither is a mechanism a firm applies to [7, 8].

Two well suited, three partly suited, two less suited. The two well-suited rows are public-goods data and infrastructure, and mission-driven public-good science, and both are well suited for the same structural reason: every one of the twelve milestone artifacts in §5 is deposited publicly by plan, and the programme has one mission with a stop condition and a closure budget.

A firm claiming to be well suited to all seven would be claiming to be a university, a corporate laboratory, a federal laboratory and a focused research organization at once. The report's own table exists to make that claim impossible to state.

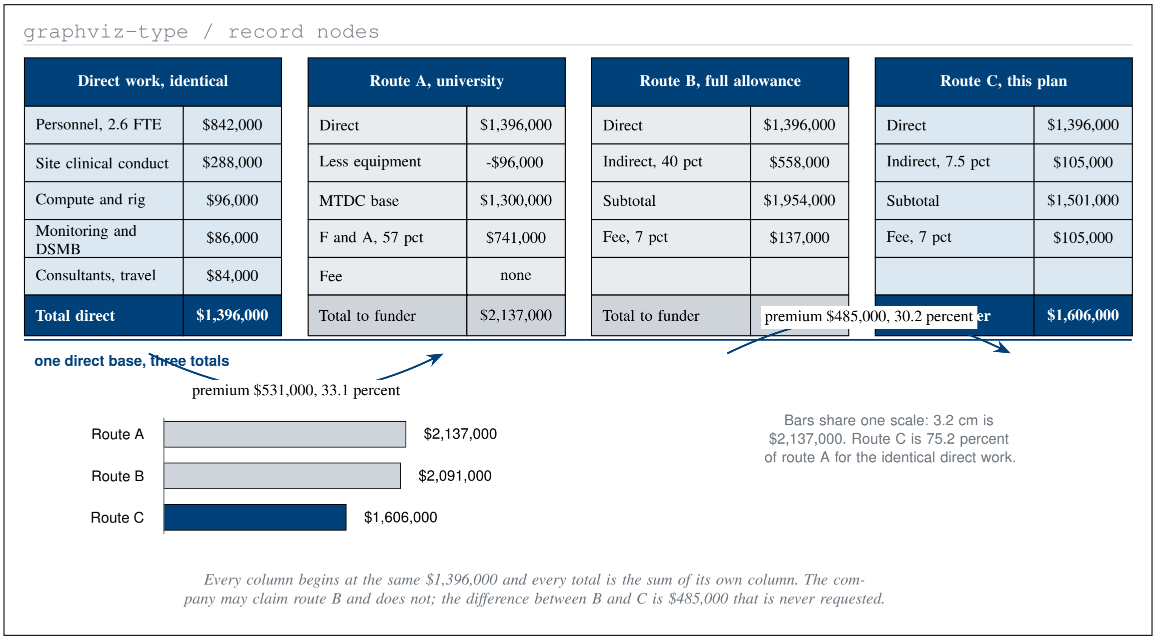
d2-type / true grid

Activity	University	Corporate lab	Federal lab	New institutions	SBIR firm 2.6 FTE
Curiosity driven, investigator led	Well suited	Less suited	Partly suited	By design	Partly suited
Larger scale, engineering intensive	Less suited	Partly suited	Partly suited	By design	Less suited
Long horizon platform and tools	Less suited	Partly suited	Well suited	By design	Partly suited
Public goods data and infrastructure	Partly suited	Less suited	Partly suited	By design	Well suited
Mission driven, public good science	Partly suited	Less suited	Partly suited	By design	Well suited
Proprietary product development	Less suited	Well suited	Less suited	By design	Partly suited
Workforce training and apprenticeship	Well suited	Partly suited	Less suited	By design	Less suited
					2 well, 3 partly, 2 less

Well or partly suited, the report's own scoring — Less suited — Well suited, this company

The rule at the right marks where the report's own columns end and the column this paper adds begins. The four left-hand columns are the report's scoring restated in words, because its plus, tilde and cross glyphs are not legible at this measure.

The report's own seven activities, its four institutional forms, and a fifth column it does not carry. An SBIR firm of 2.6 people is well suited to two of the seven and unsuited to two others.



The same 1,396,000 of direct work priced three ways. A university F and A rate at 57 percent adds 741,000, the full SBIR allowance adds 695,000, and the rate this plan claims adds 210,000 in all.

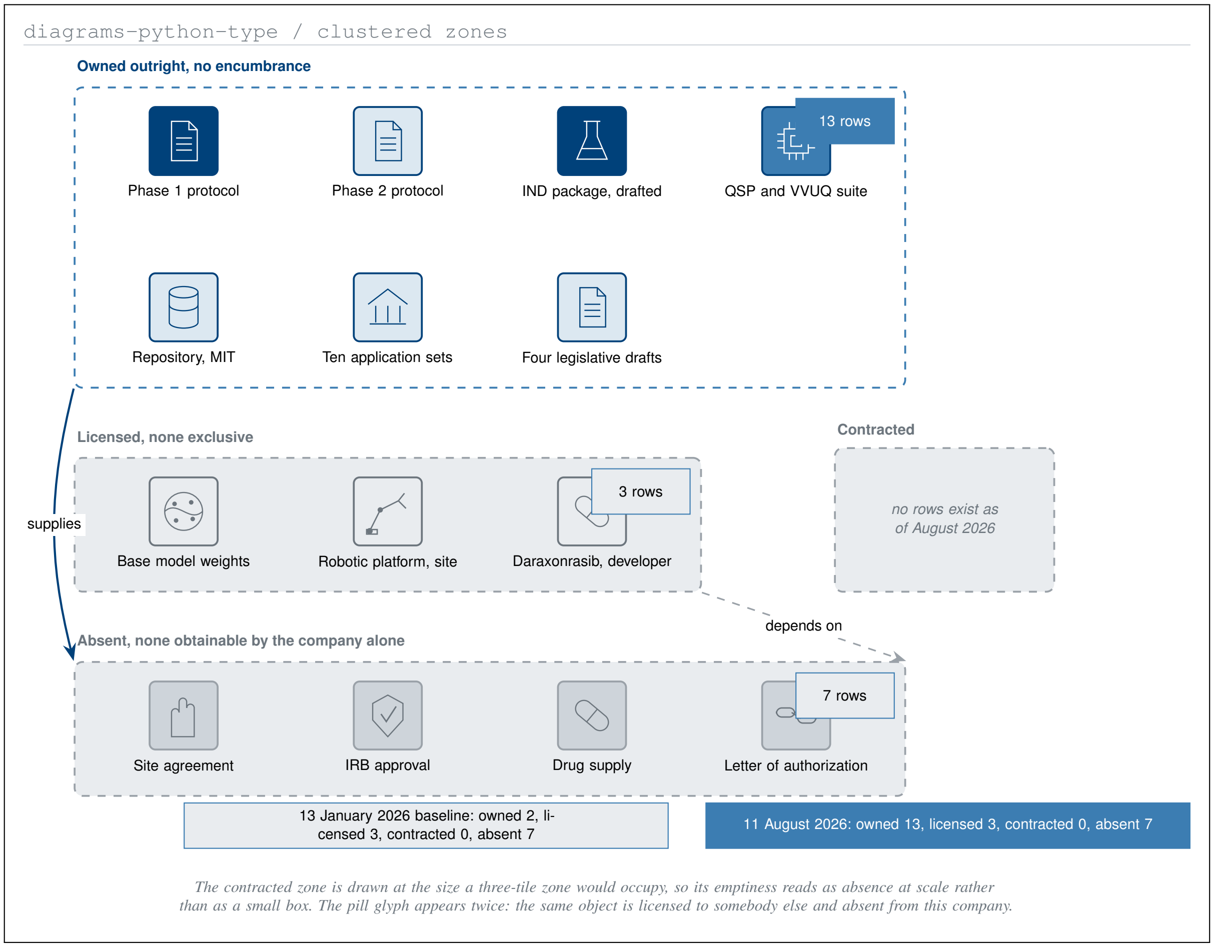
1.4 The Clause That Fits, and What It Is Worth

ChemicalQDevice has no negotiated indirect-cost rate agreement, no facilities to recover, and no administrative layer to fund. That is a structural fact about a 2.6 person firm rather than a concession, and it has a price a program officer with a fixed appropriation can read directly.

Line	Route A, university	Route B, full allowance	Route C, this plan
Direct costs	$1,396,000	$1,396,000	$1,396,000
Equipment removed from base	-$96,000	not applicable	not applicable
Overhead base	$1,300,000 MTDC	$1,396,000 direct	$1,396,000 direct
Overhead rate	57 percent	40 percent	7.5 percent
Overhead	$741,000	$558,000	$105,000
Fee at 7 percent	none	$137,000	$105,000
Total to the funder	$2,137,000	$2,091,000	$1,606,000
Premium over route C	$531,000, 33.1 pct	$485,000, 30.2 pct	zero

The same direct work under three overhead regimes. The 57 percent rate is a typical negotiated university rate; the 40 percent is the SBIR allowance available without a rate agreement; the 7.5 percent is what this firm documents.

The 7.5 percent rate covers insurance, audit and accounting, and nothing else, because there is nothing else to cover [9]. The company could claim the 40 percent allowance without documentation and does not [10]. For a program officer holding a fixed appropriation the consequence is arithmetic rather than rhetorical: the same trial, run by the same person, costs $531,000 less than it would through an institution and $485,000 less than this mechanism permits.



Four zones and thirteen tiles. The owned zone holds everything the company made; the contracted zone is drawn at full size and holds nothing, because an empty zone is the finding this figure reports.

2 The Entity and the Asset

2.1 What the Entity Is

ChemicalQDevice is a California corporation. Its sole officer, sole director and sole owner is Kevin Kawchak. It has no employees at the date of writing, no negotiated indirect-cost rate agreement, and no facilities of its own. It is therefore more than 50 percent directly owned and controlled by one United States individual, which is the eligibility test the SBIR programme applies [11], and §4 is written to keep it so through the thirty-three months this plan covers.

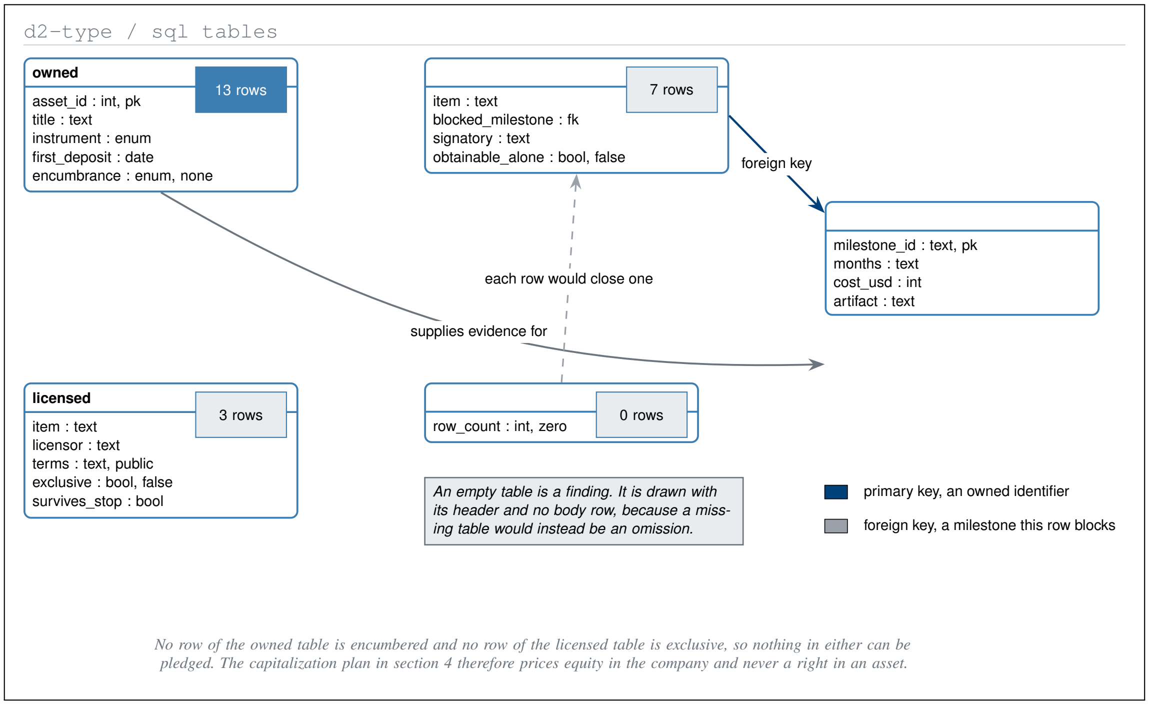
2.2 Owned

Asset	Instrument	First deposit	Identifier
Phase 1 protocol	Zenodo	2026-06-21	10.5281/zenodo.20780121
Phase 2 protocol	Zenodo	2026-06-24	10.5281/zenodo.20807027
IND package, drafted, not filed	Zenodo	2026-07-01	10.5281/zenodo.21097442
Phase 1 LLM document guidance	Zenodo	2026-06-29	10.5281/zenodo.21018646
PI LLM adoption guide	Zenodo	2026-06-26	10.5281/zenodo.20843290
QSP simulation stack and VVUQ suite	Zenodo	2025-08	10.5281/zenodo.17001137
Daraxonrasib identification meta-analysis	Zenodo	2025-06	10.5281/zenodo.15735068
Funding application v1.0	Zenodo	2026-07-07	10.5281/zenodo.21232965
Funding application v2.0	Zenodo	2026-07-12	10.5281/zenodo.21317266
Patient robot advocacy paper	Zenodo	2026-07-31	10.5281/zenodo.21720120
H. R. 9510 bill v5.0 and companions	Zenodo	2026-06-12	10.5281/zenodo.20619762
Repository, MIT licensed	GitHub	2026-08	`physical-ai-oncology-trials` v4.5.0
Ten funding application file sets	Repository	2026-08-04	`funding/pdac-funding-applications`

Thirteen owned assets, every one deposited and addressable. No row carries an encumbrance, no row is subject to a licence granted to a third party, and no row was produced under a grant that reserves rights in it.

What the thirteen rows cost to produce is on the record and is unusual enough to state. The empirical triplicate, the ten-arm quantitative systems pharmacology virtual trial, and the digital-twin platform were each produced for an estimated \$36,330 and in about one month, against industry benchmarks of above \$120,000, above \$2,000,000, and \$28,000 to \$700,000 respectively [12]. Those figures are estimates from the source set and are reported as such, not as audited accounts; Table 17 carries them with their benchmarks.

The consequence for a funder is that none of the thirteen appears in the budget. The protocol is written, the simulation stack is built, the verification suite is frozen, and the IND package is drafted [13, 14, 15, 16]. Four further rows are documents about how the work is run rather than about the trial itself: the two funding applications, the principal-investigator adoption guide, the large-document guidance, and the patient advocacy paper [17, 3, 18, 19, 20]. A request for \$1,606,000 is a request to run a trial, not to write one.



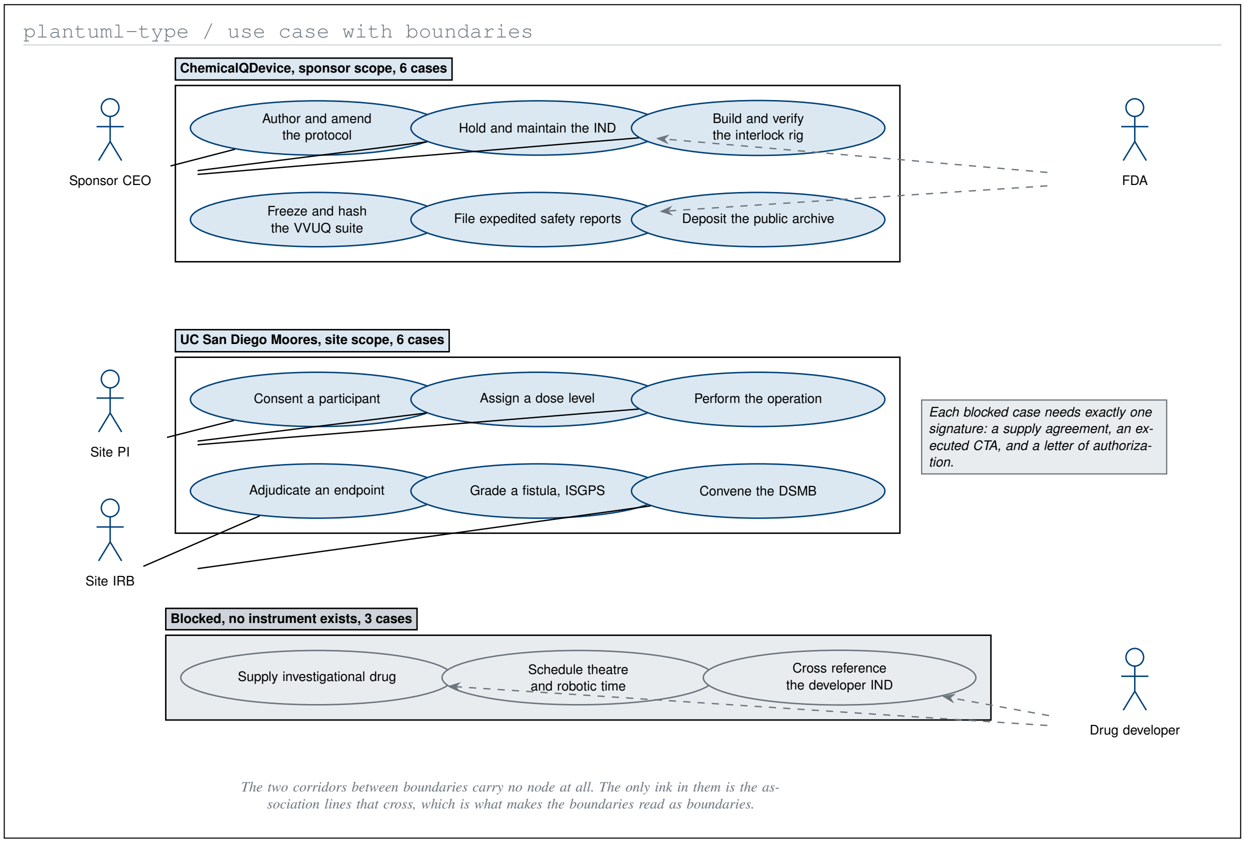
The asset register as four typed records. Thirteen rows are owned outright, three are licensed on public terms, seven are absent, and the contracted table has no rows at all as of August 2026.

2.3 Licensed, and Contracted

The four legislative drafts in the register are a separate line of work and are not on the trial's critical path: they supplied a verification vocabulary and a catalogue of clinician objections, and nothing in the protocol, the IND, or this plan is contingent on any of them being enacted [21, 22, 23, 24].

Three things are licensed and none exclusively. The base model weights are used under their vendor's public terms. The robotic platform is the site's asset and not the company's. Daraxonrasib is the developer's investigational agent, and no licence of any kind exists. None of the three survives a stop, because none is owned.

The contracted table has no rows. No clinical trial agreement, no drug supply agreement, no letter of authorization, no confidentiality agreement, no subaward, and no trial-specific insurance binder is in force.



Two system boundaries, twelve use cases, and the corridor between them holding three that neither party may execute today. Each of the three is blocked by exactly one missing signed instrument.

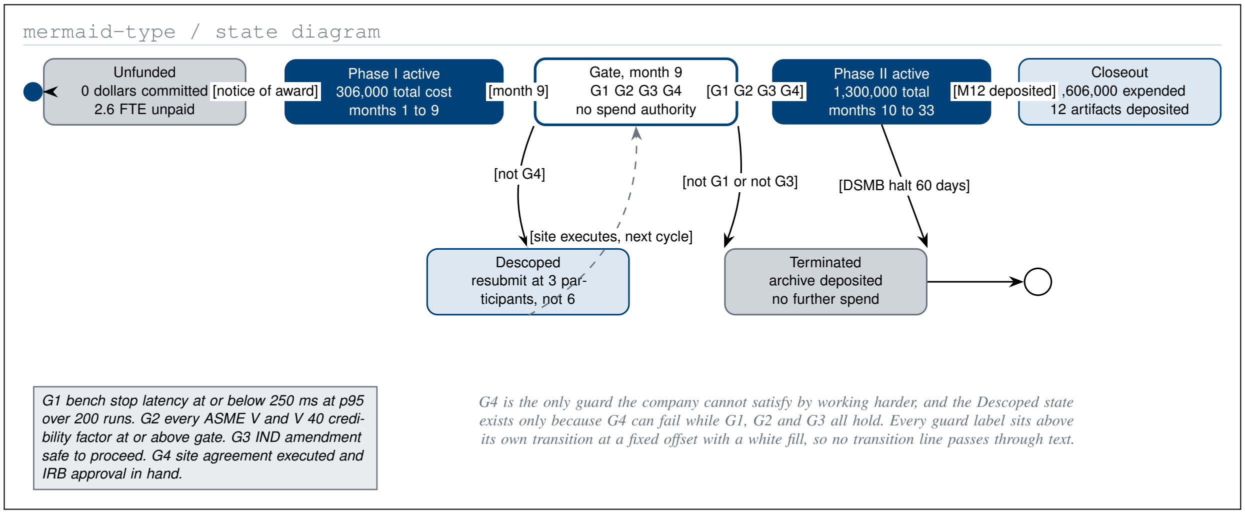
2.4 Absent

Instrument absent	Blocks	Only signatory	Obtainable by the company alone
Site agreement, executed	M1	UC San Diego	No
IRB approval	M2	Site institutional review board	No
Drug supply agreement	M6	Revolution Medicines	No
Letter of authorization, IND cross-reference	M5	Revolution Medicines	No
Theatre and robotic platform time	M7	UC San Diego	No
Trial-specific liability insurance	M6	Carrier	No, needs an executed CTA
A second full-time employee	M3	Funded by the Phase I award	No, needs the award

Seven absent instruments, the milestone each blocks, and the single party that can supply it. The last column reads no on every row, which is the whole content of the table and the reason §8 exists.

Three use cases sit in the corridor, and each is blocked by exactly one signature. No amount of company work removes any of the three, which is the honest reading of the register and the reason the twelve milestones in §5 open with a site feasibility memorandum rather than with an experiment.

The register is dated. On 13 January 2026, when the first proposal was released, the owned table held two rows and the absent table held the same seven [25]. In the seven months since, the owned table has grown by eleven rows and the absent table has not lost one. That asymmetry is the single most useful thing in this section: it says precisely which kind of work the author can do alone and which kind he cannot.



One award in five states, and the four guards that must all hold at month nine. Two of the four are technical, one is regulatory, and the fourth belongs to an institution the company cannot bind.

3 The $1.6M Gate and the $3.5M Programme

3.1 One Budget, Reused Without Re-derivation

The five-year frame is taken verbatim from the funding application of 12 July 2026 and is not re-derived here: $700,000 per year of direct cost, $3,500,000 over five years, no cost share, no programme income anticipated [3]. It divides into four layers of $1,600,000 for clinical conduct, $720,000 for IND maintenance and safety reporting, $780,000 for the interlock rig with its logging and audit replay, and $400,000 for the verification package and archive.

3.2 The Award, and the Work Inside It

An SBIR award is a total-cost figure and not a work figure. It carries direct costs, indirect costs, and a fee that is calculated on the sum of the two and is not itself a cost of the project [26]. A milestone table that confuses the two will not survive a first reading.

Component	Phase I, 9 months	Phase II, 24 months	Both, 33 months
Direct costs	$266,000	$1,130,000	$1,396,000
Indirect at 7.5 percent of direct	$20,000	$85,000	$105,000
Subtotal	$286,000	$1,215,000	$1,501,000
Fee at 7 percent of subtotal	$20,000	$85,000	$105,000
Total award	$306,000	$1,300,000	$1,606,000
Overhead and fee load on direct	15.0 percent	15.0 percent	15.0 percent

The two awards decomposed. A $1,606,000 award carries $1,396,000 of direct work at a 15.0 percent load, against an SBIR indirect allowance of 40 percent of direct costs that this plan does not claim.

G1, bench stop latency. Two hundred instrumented runs on the interlock rig must return a 95th-percentile stop latency at or below 250 ms, against the protocol's arm-stop requirement of 3 ms and system-stop requirement of 500 ms. The company can satisfy this alone, and M3 pays for it.

G2, verification credibility. Every credibility factor in the verification suite must sit at or above its ASME V&V 40 gate, and the suite must be hashed so the state it passed in is the state that is archived [27, 28]. The company can satisfy this alone, and M4 pays for it.

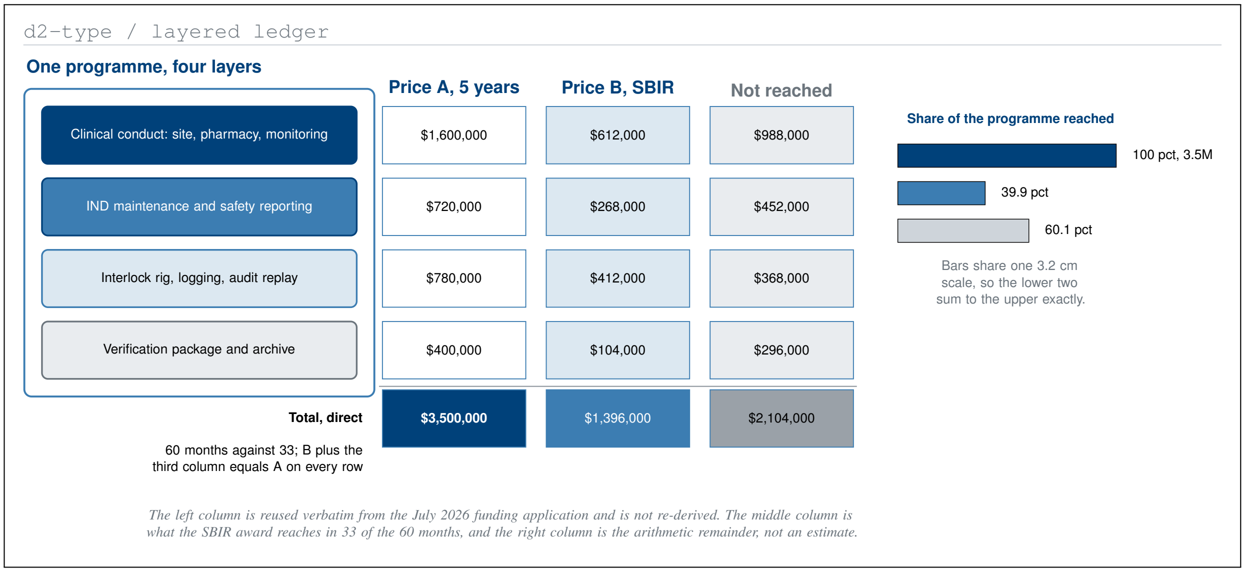
G3, regulatory. The IND amendment must reach safe to proceed with the 30-day clock closed and no clinical hold [29]. The company files it; the agency decides it.

G4, institutional. The site agreement must be executed and IRB approval in hand. This is the only guard no amount of company work can satisfy, and it is why the Descoped state exists at all: G1, G2 and G3 can all hold while G4 fails, and the correct response to that combination is a smaller Phase II rather than a termination.

3.3 Phase I, Nine Months, $306,000

Line	Amount	What it buys
CEO and sponsor lead, 0.6 FTE, 9 months	$94,500	Protocol, IND amendment, site negotiation
Verification engineer, 0.5 FTE, 9 months	$63,000	Rig build, instrumentation, VVUQ freeze
On-premises compute, two GPU nodes	$38,000	Simulation and replay capacity, amortized
Interlock rig hardware and instrumentation	$31,000	The bench that G1 is measured on
Regulatory preparation and FDA package	$27,500	The amendment G3 turns on
Site feasibility, budget, IRB fees	$24,000	M1 and M2
Travel and dissemination	$8,000	One programme review, one deposit
Subtotal, direct	$266,000	
Indirect at 7.5 percent and fee at 7 percent	$40,000	Insurance, audit, accounting, and the fee
Total award	$306,000	

Phase I in eight direct lines plus the load. No participant is consented and no drug is dispensed inside Phase I: every dollar buys either an instrument, a verification, or a signature.



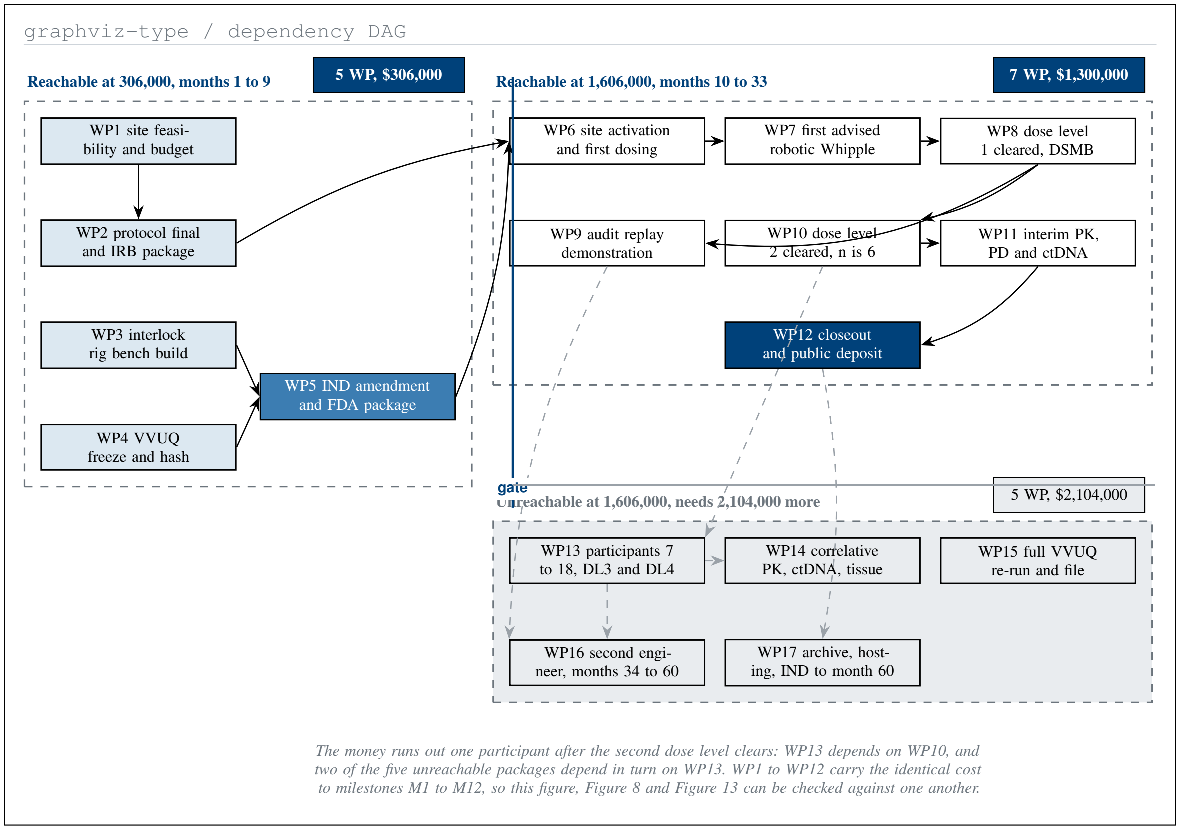
The left column is reused verbatim from the July 2026 funding application and is not re-derived. The middle column is what the SBIR award reaches in 33 of the 60 months, and the right column is the arithmetic remainder, not an estimate.

One programme, four layers, and two prices for the same work. The SBIR route reaches 1,396,000 of direct effort inside a 1,606,000 award; the third column is the 2,104,000 that it does not reach.

3.4 Phase II, Twenty-Four Months, $1,300,000

Line	Amount	What it buys
CEO and sponsor lead, 0.8 FTE, 24 months	$280,000	Sponsor obligations, safety reporting
Verification engineer, 1.0 FTE, 24 months	$224,000	Logging, audit replay, build signing
Clinical research coordinator, 0.5 FTE	$96,000	Case report forms, deviations, queries
Site clinical conduct, 6 participants	$288,000	$48,000 per participant, all visits
Pharmacy, specimens, central laboratory	$104,000	Dispensing, handling, assay
Independent monitoring and DSMB	$86,000	Source verification and four DSMB sittings
IND maintenance and safety reporting	$72,000	Annual report, expedited reports
Audit replay, logging and archive build	$98,000	M9 and the deposit at M12
Consultants, biostatistics and regulatory	$34,000	Fixed fee, not tied to outcome
Travel, dissemination and closeout	$18,000	Two reviews, one public deposit
Subtotal, direct	$1,130,000	
Indirect at 7.5 percent and fee at 7 percent	$170,000	Insurance, audit, accounting, and the fee
Total award	$1,300,000	

Phase II in ten direct lines plus the load. Six participants at $48,000 each is the largest single line; if the site executes late, Phase II is resubmitted for three participants, which is the Descoped state of Figure 7.



Seventeen work packages and every edge between them. Five close at 306,000, seven more at 1,606,000, and five are unreachable at that figure, all five downstream of the sixth participant.

3.5 The Same Work, Priced Twice

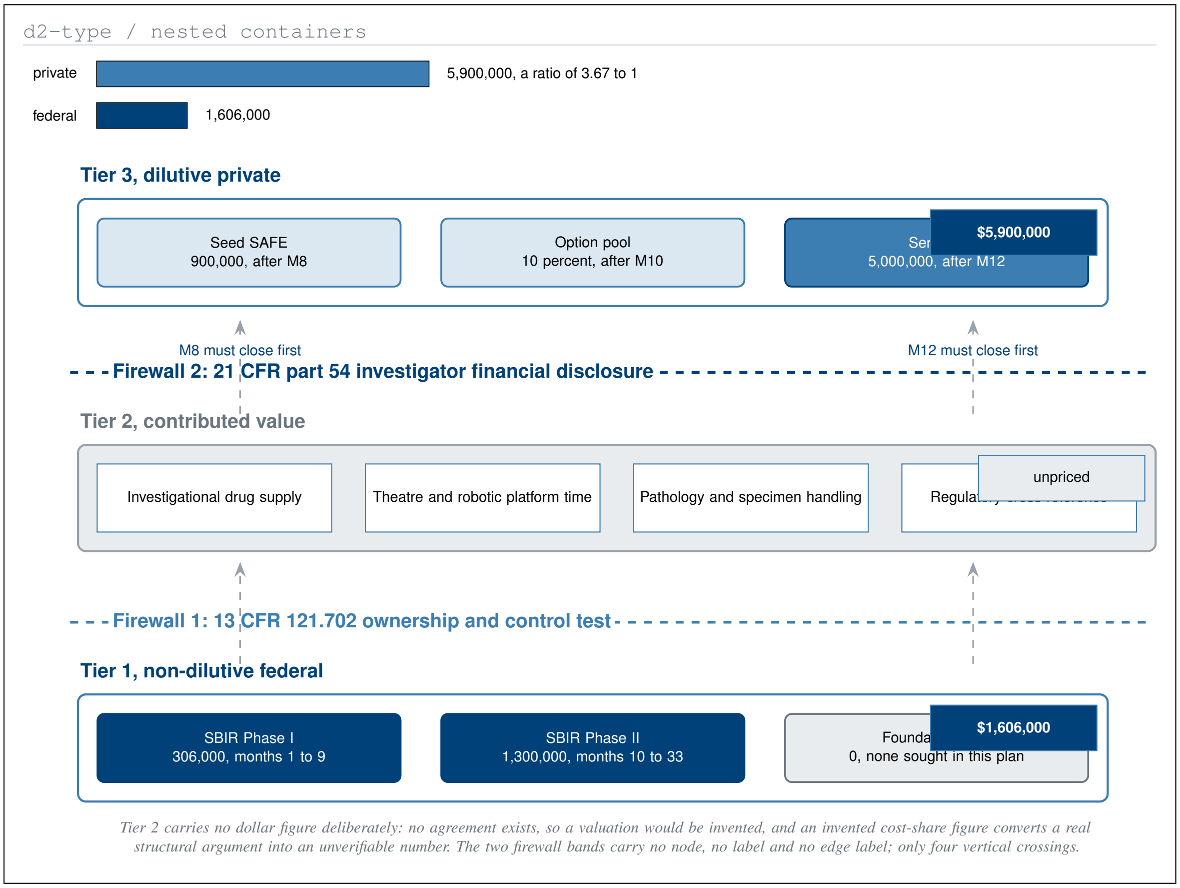
Layer	Price A, 5 years	Inside the award	Not reached
Clinical conduct: site, pharmacy, monitoring	$1,600,000	$612,000	$988,000
IND maintenance and safety reporting	$720,000	$268,000	$452,000
Interlock rig, logging, audit replay	$780,000	$412,000	$368,000
Verification package and archive	$400,000	$104,000	$296,000
Total, direct	$3,500,000	$1,396,000	$2,104,000
Term	60 months	33 months	27 months

The four-layer budget under two prices. Every row reconciles: the middle column plus the right column equals the left, and the three totals are the sums of their own columns.

The delta is not a shortfall to apologise for. It is the price of the mechanism, and it decomposes into five identifiable things: twenty-seven additional months of term, participants seven through eighteen, two further dose levels and the expansion at the recommended Phase 2 dose, the correlative pharmacology, and the full verification re-run with the permanent archive.

3.6 What the Money Reaches

Two of the five unreachable packages are not about participants at all. WP15, at $228,000, is the full verification re-run against post-trial data and the credibility file that goes with it; without it the credibility score of 81.9 remains a pre-trial claim. WP17, at $298,000, is the permanent archive, the replay hosting, and IND maintenance out to month 60. WP17 is the single package whose removal changes what §10 is able to claim, which is why it sits in the unreachable cluster rather than being quietly absorbed into WP12.



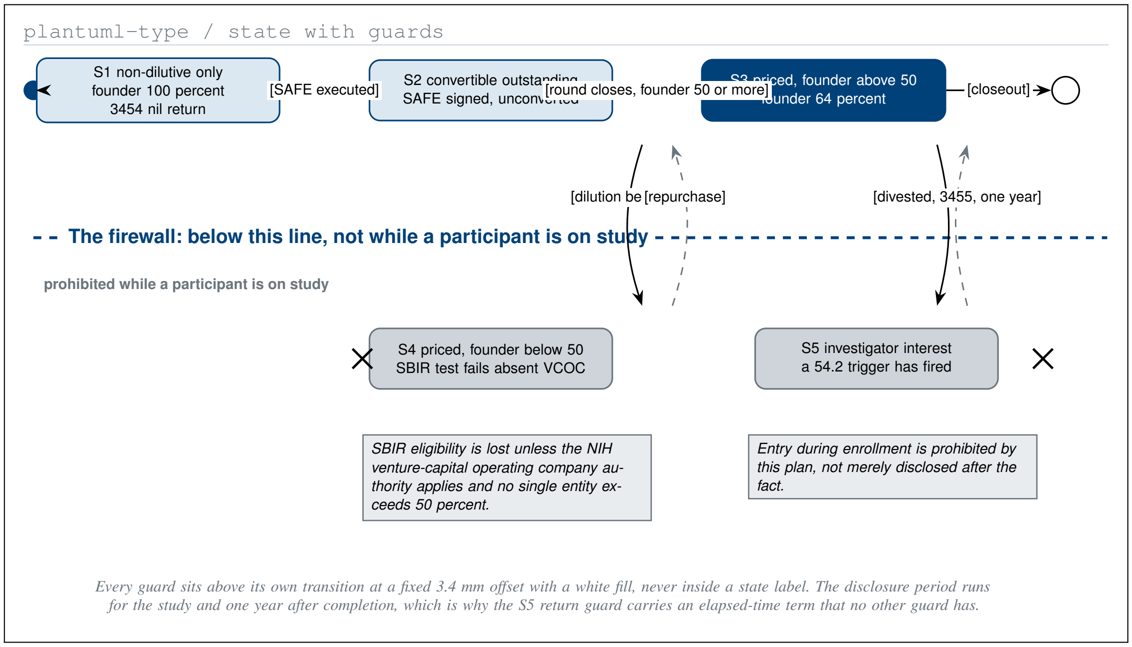
Three capital tiers and the two gaps between them. Federal money at the base, contributed value unpriced in the middle, private capital above, and 5,900,000 against 1,606,000 is 3.67 to one.

4 Non-Dilutive to Dilutive Bridge

4.1 Three Tiers of Capital

The plan holds $1,606,000 of non-dilutive federal money, an unpriced tier of contributed drug, theatre, pathology and regulatory support, and $5,900,000 of dilutive private capital raised only after a milestone closes. On cash alone that is 3.67 to one, against the FY 2028 annexed instruction that agencies target “at least a 3:1 leverage of private to Federal investment” [4]. The ratio is reached before any contributed value is counted, which is the argument for counting it at zero.

Tier 2 carries no dollar figure, for the reason the parent work gives and this one does not improve on: no agreement exists with any developer or site, so a valuation would be invented, and an invented cost-share figure converts a real structural argument into an unverifiable number. The claim made about tier 2 is qualitative and checkable, that four named categories of cost do not appear in the federal request.



The capital firewall as five states. Two of them, below the rule, may not be entered while a participant is on study. Every guard names its own regulation and the form the entry would require.

4.2 The Capital Firewall

The author is chief executive of the sponsor and holds 100 percent of it. If he were also a clinical investigator on the trial, every trigger in 21 CFR §54.2 would fire at once [30]. The plan therefore assigns the investigator role entirely to the site: the operating surgeon and the principal investigator are the site's employees, hold no ChemicalQDevice equity, receive no compensation tied to study outcome, and each files Form FDA 3454 with a nil return [31].

Trigger	Citation	Threshold	Form
Compensation affected by study outcome	21 CFR §54.2(a)	Any amount	3455
Significant equity, sponsor not publicly traded	21 CFR §54.2(b)	Any equity interest	3455
Significant equity, sponsor publicly traded	21 CFR §54.2(b)	Above $50,000	3455
Proprietary interest in the tested product	21 CFR §54.2(f)	Any patent, licence or royalty	3455
Significant payments of other sorts	21 CFR §54.4(a)(3)(ii)	Above $25,000 beyond study cost	3455
No disclosable interest	21 CFR §54.4(a)(1)	Certification by the sponsor	3454

The five disclosure triggers and the certification that applies when none has fired. The disclosure period runs for the duration of the study and one year after its completion, which is why divestment alone does not clear S5.

4.3 What the Firewall Actually Separates

Dose assignment. A Phase 1 3+3 escalation carries no randomization, so the firewall cannot bind on a randomization schedule. It binds instead on dose assignment, which is the Phase 1 analogue: the level a participant receives is assigned by the site principal investigator on the recommendation of an independent data and safety monitoring board. The wording is carried forward unchanged so that it still binds when the Phase 2 successor introduces randomization [15].

Endpoint adjudication. Whether an event is a dose-limiting toxicity and what grade it carries is adjudicated at the site against CTCAE and, for pancreatic fistula, against the ISGPS grading [32]. No sponsor employee sits on the adjudication.

The analysis plan. The statistician is contracted at a fixed fee that is not tied to outcome, and the analysis populations and interval method are locked before the first participant is dosed. A locked plan is what makes the fixed fee meaningful; either alone would not be enough.

4.4 The Capitalization Table

Stage	Instrument and gate	Cash raised	Founder holding	13 CFR 121.702 test
0	Founder common, 8,000,000 shares	none	100.0 percent	Passes
1	Seed SAFE, after M8 clears	$900,000	88.9 percent	Passes
2	Option pool, 10 percent, after M10	none	80.0 percent	Passes
3	Series A, after M12 deposits	$5,000,000	64.0 percent	Passes
4	Series B, beyond this plan	not budgeted	46.5 percent	Fails absent VCOC

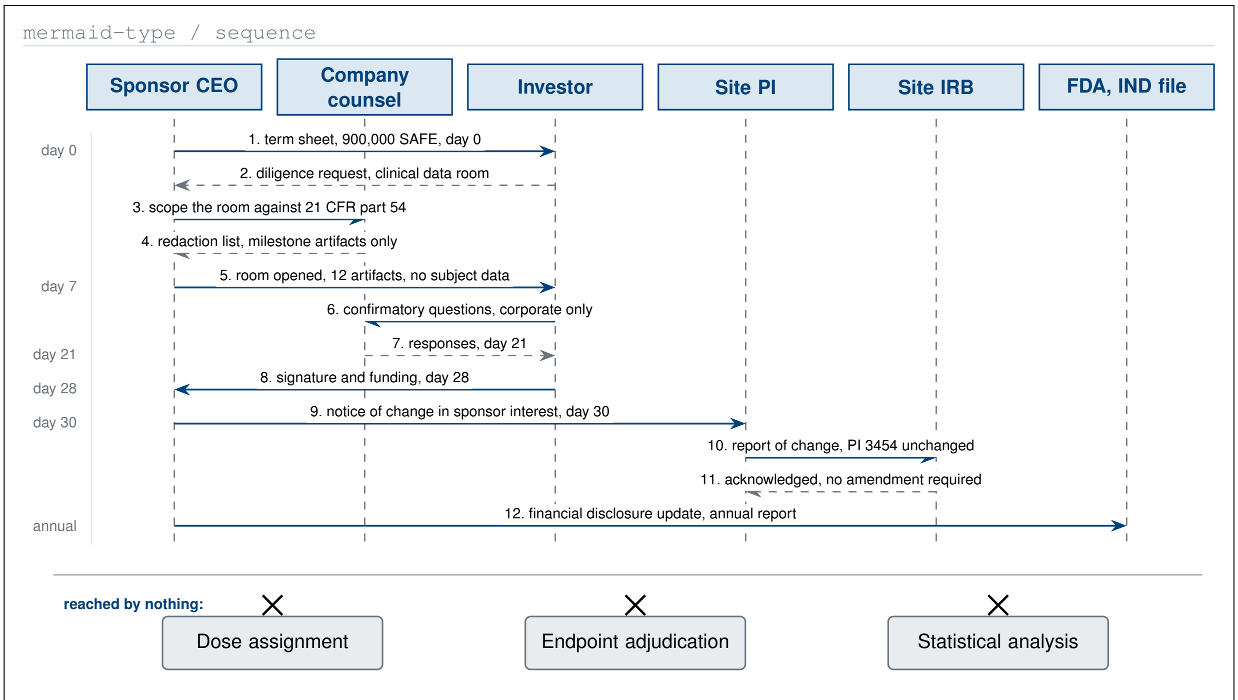
Five capital stages and the ownership test at each. Stages 0 to 3 fall inside the thirty-three months this document covers and all four pass. Stage 4 is shown because a plan that hid it would be less useful, not more.

Stage 4 is the one that fails. A Series B at the size a Phase 2 would require takes the founder below 50 percent, and SBIR eligibility is then lost unless the venture-capital operating company authority applies and no single such entity holds a majority [11, 33]. The plan's answer is not a structuring trick: stage 4 sits outside the thirty-three months this document covers, and no Phase II obligation extends into it.

4.5 The Sequence

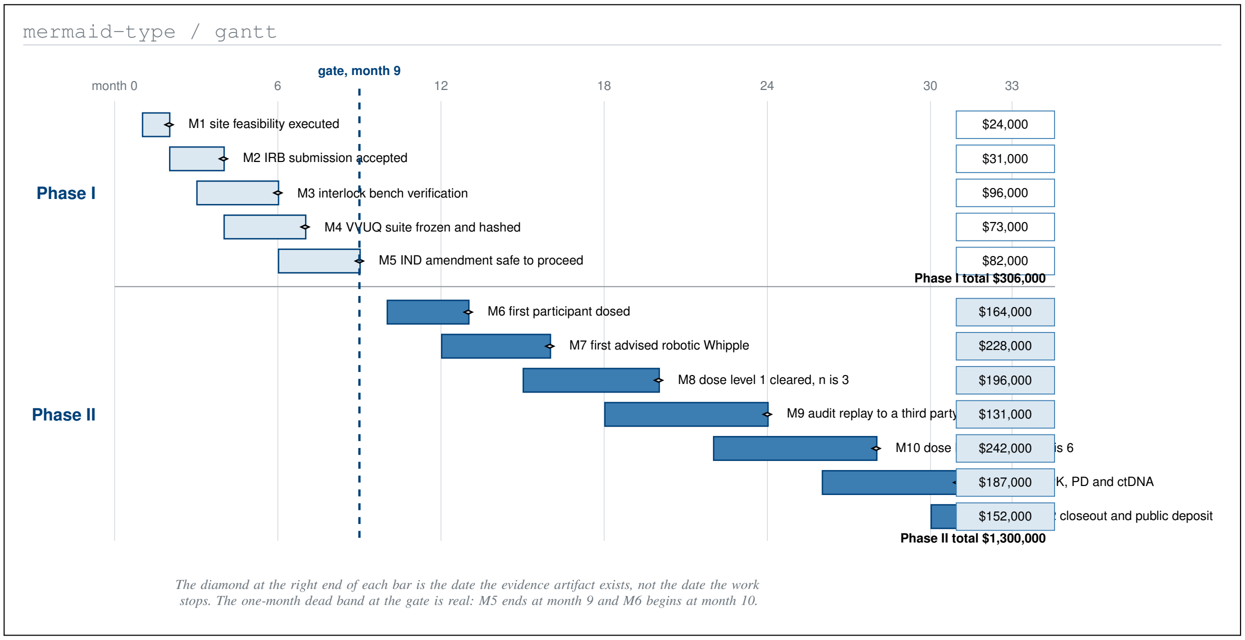
Two dates in that sequence are obligations rather than courtesies. The notice to the site principal investigator goes out within thirty days of any change in the sponsor's financial position, and the IND financial disclosure commitment is updated at the next annual report [29]. Neither is triggered by the size of the round; both are triggered by the fact of it.

Every tier 3 instrument is gated behind a milestone: the seed instrument behind M8, the option pool behind M10, and the Series A behind M12. No private capital enters the entity before dose level 1 has cleared, which is the single structural commitment this section exists to make.



A 3+3 escalation has no randomization, so the firewall binds on dose assignment rather than on a schedule. The three struck lifelines are decided by the site principal investigator, by the site against CTCAE and ISGPS, and by a fixed-fee statistician working to a plan locked before first dose.

Eleven messages during a financing round, and the three lifelines no message reaches. Dose assignment, endpoint adjudication, and the analysis plan are untouched between term sheet and signature.



Thirty-three months, twelve milestones, and the artifact date on each. Phase I holds five at 306,000 dollars; Phase II holds seven at 1,300,000. The gate at month nine is the only hard boundary.

5 Twelve Milestones a Program Officer Can Audit

5.1 What an Auditable Milestone Is

A milestone is auditable only if it carries four things: an evidence artifact that exists as a file, a cost that sums into its phase, a stop condition stated before the work begins, and a named funder mechanism that pays for it. All twelve rows below carry all four. The five Phase I rows sum to $306,000 and the seven Phase II rows sum to $1,300,000, both exactly, and a milestone table that does not sum to its award is the fastest way to lose a reader who has read several hundred of them.

5.2 The Twelve Rows

ID	Months	Milestone	Cost	Phase
M1	1 to 2	Site feasibility executed	$24,000	Phase I
M2	2 to 4	IRB submission accepted	$31,000	Phase I
M3	3 to 6	Interlock rig bench verification	$96,000	Phase I
M4	4 to 7	VVUQ suite frozen and hashed	$73,000	Phase I
M5	6 to 9	IND amendment safe to proceed	$82,000	Phase I
M6	10 to 13	First participant dosed	$164,000	Phase II
M7	12 to 16	First advised robotic Whipple completed	$228,000	Phase II
M8	15 to 20	Dose level 1 cleared, three participants	$196,000	Phase II
M9	18 to 24	Audit replay demonstrated to a third party	$131,000	Phase II
M10	22 to 28	Dose level 2 cleared, six participants	$242,000	Phase II
M11	26 to 31	Interim PK, PD and ctDNA readout	$187,000	Phase II
M12	30 to 33	Closeout and public deposit	$152,000	Phase II

The twelve milestones with their months and cost. M1 to M5 sum to $306,000 and M6 to M12 sum to $1,300,000; the same twelve costs appear as work packages WP1 to WP12 in Figure 9.

ID	Evidence artifact	Stop condition	Mechanism
M1	Countersigned feasibility memorandum	No site by month 4; Phase I withdrawn	SBIR Phase I
M2	IRB acknowledgment and protocol v1.0	Two rejections; protocol re-scoped	SBIR Phase I
M3	200-run latency report with p95 table	p95 above 250 ms; no first-in-human	SBIR Phase I
M4	VVUQ report and SHA-256 manifest	Any credibility factor below its gate	SBIR Phase I
M5	FDA acknowledgment, 30-day clock closed	Clinical hold; Phase II not requested	SBIR Phase I
M6	Consent, dispensing record, day-1 CRF	No enrollment by month 15; descope to three	SBIR Phase II
M7	Operative note, advisory log, replay bundle	Any advisory override event; advisory read-only	SBIR Phase II
M8	DSMB minutes and safety table	Two of three DLTs; de-escalate	SBIR Phase II
M9	Replay record and hash-match certificate	Any non-reproducible record; hold enrollment	SBIR Phase II
M10	Cumulative safety table, ISGPS grades	Grade B or C fistula above 30 percent	SBIR Phase II
M11	PK report and ctDNA clearance table	No target engagement at dose level 2	SBIR Phase II
M12	Zenodo deposit, report, repository tag	None; this is the terminal artifact	SBIR Phase II

The same twelve rows keyed by ID, carrying what proves each milestone, what halts it, and which mechanism pays. Every stop condition is a number or a signed instrument; none is a judgement call.

M6 additionally requires that the trial be registered before the first participant is enrolled, and the registration record is part of that milestone's artifact set [34].

Everything in the $2,104,000 delta of §3 lies beyond M12 and carries no milestone row here, deliberately. A milestone schedule that promised work no award pays for would not be auditable.

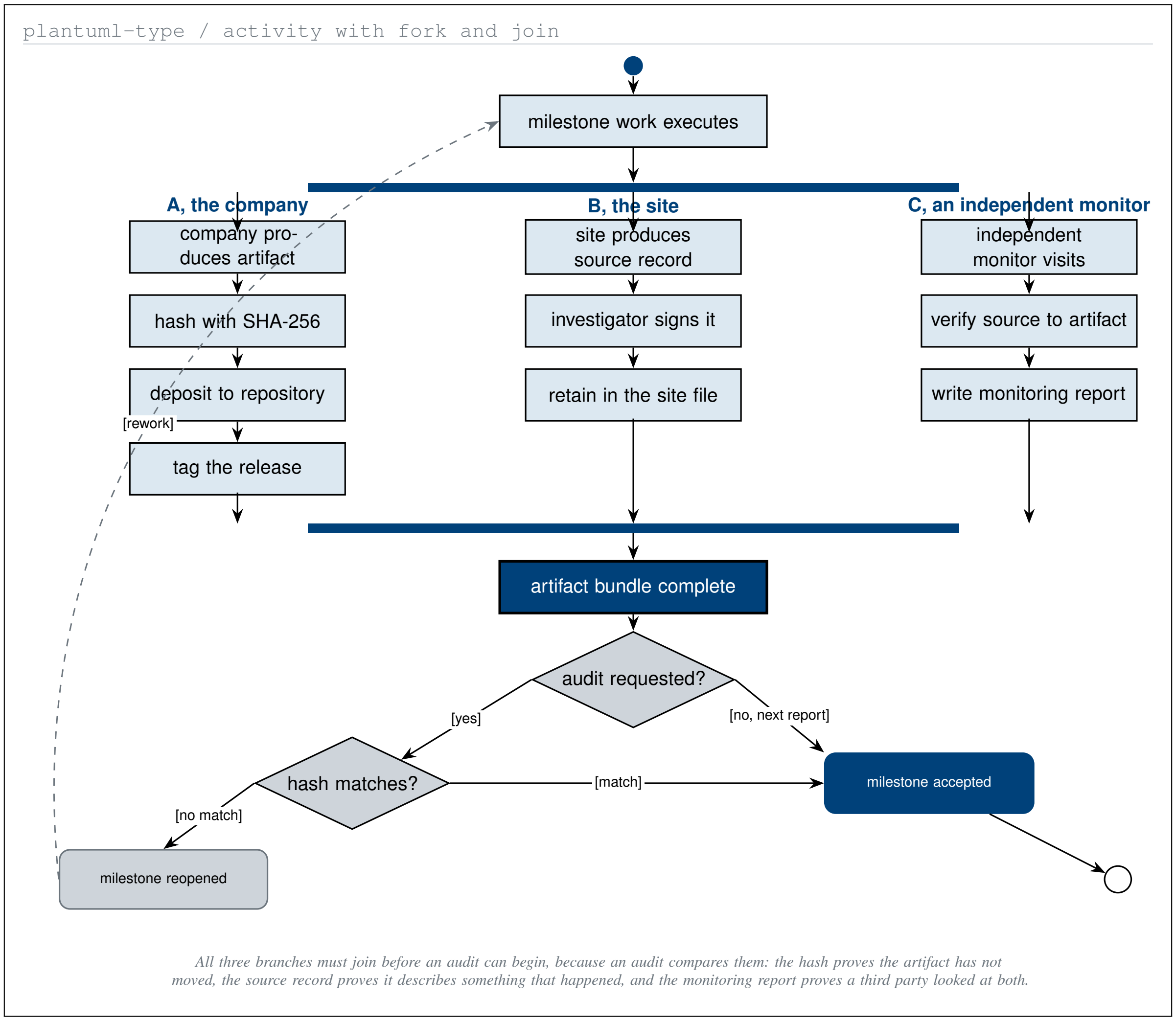
5.3 How a Milestone Is Closed

Any one branch alone is a claim rather than evidence. A hash with no source record proves that a file has not changed and says nothing about whether it describes an event. A source record with no hash proves an event occurred and says nothing about whether the artifact still matches it. The monitoring report is what joins the two, and it is written by somebody the company does not employ [35].

5.4 How the Programme Stops

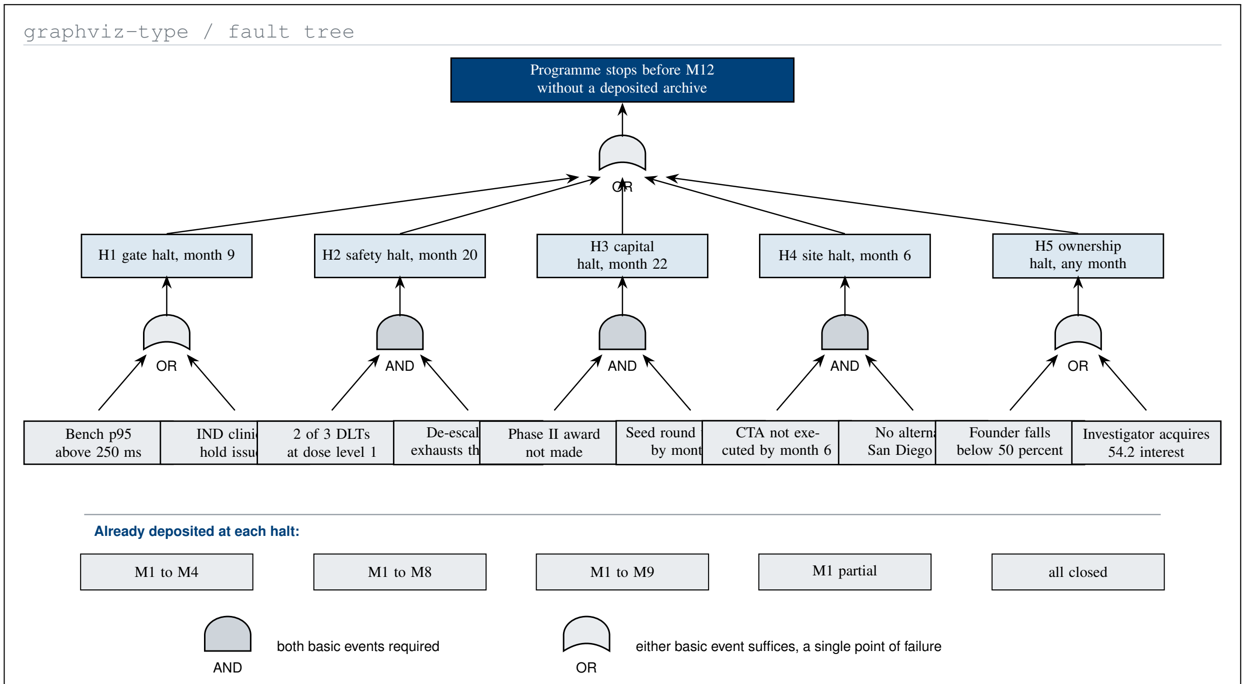
The two OR branches are the honest part of the figure. H1 fails on either a bench latency above 250 ms at the 95th percentile or an IND clinical hold; H5 fails on either the founder falling below 50 percent or an investigator acquiring a §54.2 interest. Neither is technical in the sense that more engineering fixes it, and both are knowable in advance, which is the only reason they are acceptable to state on a milestone schedule.

The top event is deliberately not "the programme stops". It is a stop *without a deposited archive*, and the band beneath the tree is why the distinction matters: at every halt point the archive is already deposited for



One milestone, three concurrent evidence branches, and the join that has to close before a program officer can audit anything. The replay is the only step that can send a milestone backwards.

every milestone that has closed. Reaching the top event would require a halt plus a separate failure to deposit, and depositing is itself a Phase I deliverable that does not depend on the halt's cause. Figure 20 carries the custody arrangement that makes that true.



The top event is a stop without a deposited archive, not a stop. Three of the five branches are AND gates. The two OR branches, H1 and H5, are the only single points of failure, and neither is technical: one is regulatory and instrumental, the other is a fact about who owns the shares.

One top event, five halt points, and ten basic events below. Three are AND gates, so no single failure stops the programme. The two OR branches are the month-nine gate and ownership test.

6 The Clinical Evidence a Funder Is Buying

6.1 Six Quantities a Reviewer Can Check

Every quantity below is published or deposited, every one carries the limitation its own authors stated, and no in-silico result is presented as a clinical one. The discipline matters more here than anywhere else in the document, because a capitalization plan that overstates its evidence is asking for money under a false description.

Source	Test arm	Comparator	Tier	Limitation, in its own authors' words
RASolute 302, May 2026	13.2 mo mOS	6.6 mo mOS	Trial	Metastatic and previously treated; silent on the resectable setting
QSP simulation, 10 arms, 250 ODEs	12.8 mo mOS, HR 0.25	5.4 mo mOS	In silico	Assumes no acquired resistance and ideal pharmacodynamics; grade 3 plus high in all arms
Digital twin, 1000 patients	12.1 mo mOS	not applicable	Twin	No patient-specific PK, PD or tumour growth parameters; simple Emax model
Digital twin, PFS hazard ratio	HR 0.31	not applicable	Twin	Same source, same limitation; no immune compartments
VVUQ credibility, 55 tests	score 81.9	V and V 40 gate	Twin	A pre-trial credibility score; not a post-trial validation
Empirical triplicate, 100,000 records	G3 plus 8.0 pct	G3 plus 25.0 pct	In silico	The G12C log favours experimental while the KRAS-mutant report favours control

Six quantities with their comparator, their evidence tier, and the limitation each set of authors stated. No row can be read without the caveat that qualifies it, which is the condition on citing any of them.

In August 2025 a ten-arm quantitative systems pharmacology simulation returned median overall survival of 12.8 months against 5.4 for chemotherapy [12]. In May 2026, nine months later, RASolute 302 reported 13.2 months against 6.6 in the RAS G12 population of previously treated metastatic pancreatic cancer [36]. The simulated ratio was 2.4-fold and the observed ratio was 2.0-fold.

That proximity is a chronology observation and a hypothesis-supporting one. It is **not** a validation claim, and three differences are material: 1000 simulated against 241 enrolled; a combination against a single agent; and KRAS G12C against a primarily G12D and G12V population. Those clauses appeared in all ten prior applications in the same paragraph as the numbers, and they appear here for the same reason [1].

6.2 The Trial Being Funded

Pancreatic ductal adenocarcinoma remains the disease with the widest gap between incidence and survival in the United States [37], and the standard of care after resection is adjuvant combination chemotherapy [38]. The candidate agent was selected from a June 2025 meta-analysis across forty literature areas [39], and the 1000-patient digital twin that produced the hazard ratio of 0.31 and the credibility score of 81.9 is deposited with its verification notebooks [40].

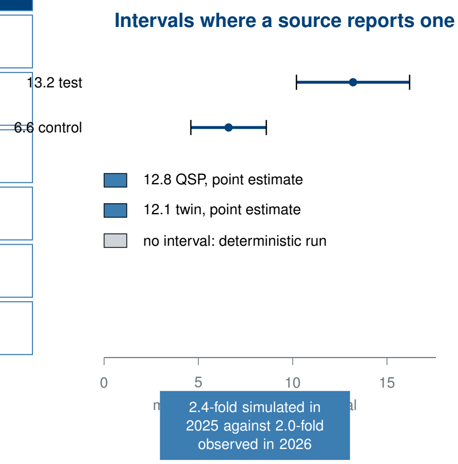
The trial is open-label, single-arm, 3+3 escalation, up to 18 treated participants, with perioperative daraxonrasib and a staged eight-arm robotic pancreaticoduodenectomy in which an on-premises large language model is confined to advisory output [14, 16]. The device specification a reviewer will ask for is fixed in the protocol: 56 degrees of freedom, 640 sensor channels, a 3 N per-arm and 18 N cumulative force cap, a 3 ms cross-arm arm stop against a 500 ms system stop, and a five-vessel no-fly gate. Dose levels are 160, 220 and 300 mg with a 28-day dose-limiting toxicity window.

d2-type / grid with interval measures

Six checkable quantities

trial · in silico or twin · peer-reviewed clinical readout

Quantity	Test arm	Comparator	Tier	Date
RASolute 302 median OS, RAS G12	13.2 months	6.6 months	Trial	2026-05
QSP simulation median OS, 10 arms	12.8 months	5.4 months	In silico	2025-08
Digital twin median OS, 1000 patients	12.1 months	not applicable	Twin	2025-10
Digital twin PFS hazard ratio	HR 0.31	not applicable	Twin	2025-10
VVUQ credibility score, 55 tests	score 81.9	V and V 40 gate	Twin	2025-10
Empirical triplicate, grade 3 plus	8.0 percent	25.0 percent	In silico	2025-07



Three quantities carry an interval and three do not. A ten-arm deterministic ODE run and a 1000-patient synthetic cohort have no sampling interval, and drawing one would invent precision the source does not claim.

Six quantities a reviewer can check against a published source, each beside its comparator and its stated limitation. Two are in silico, one is a digital twin, and three are trial or registry.

Daraxonrasib is investigational and is already in Phase 3 evaluation. It is nowhere described in this paper as first in human, and the supportable novelty claim concerns the integrated surgical and advisory workflow rather than the agent. That correction was introduced during the audit pass on the ten prior applications and is carried forward here unchanged.

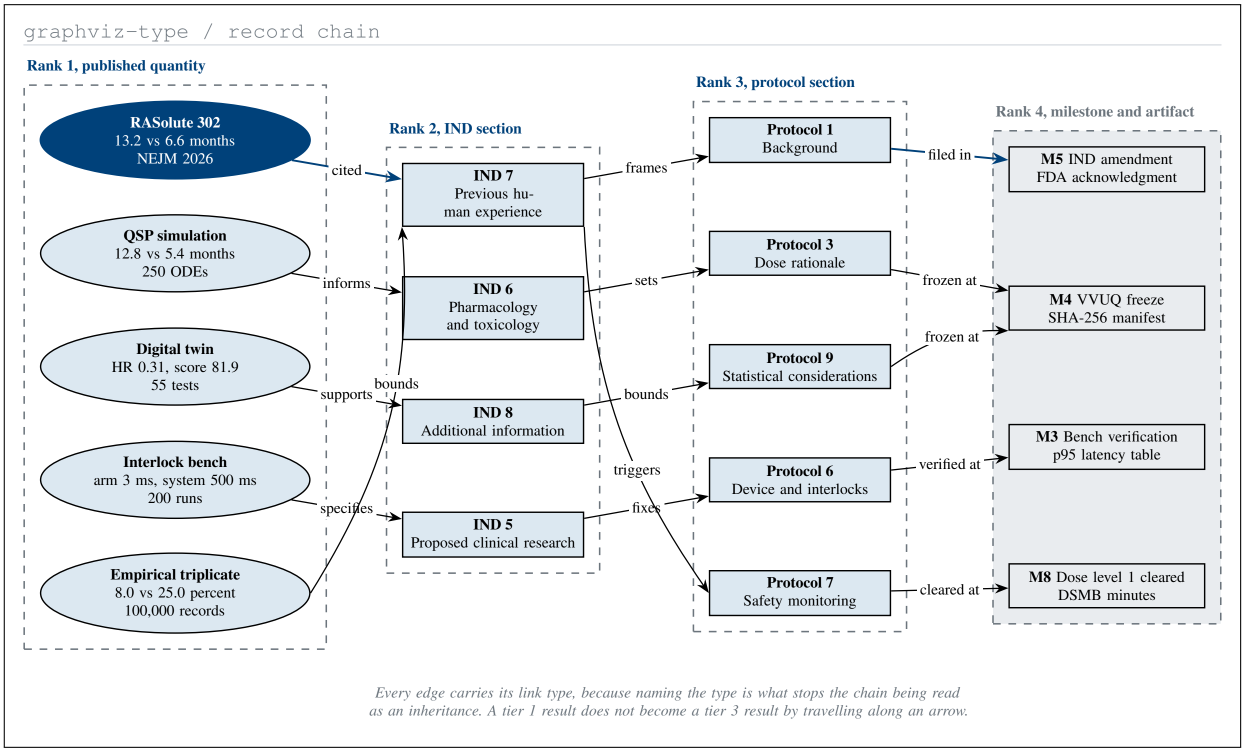
6.3 Where Each Number Lands

Five link types appear on the fifteen edges, and each licenses something different. *Cited* means a published result is quoted with its own interval and limitation, and does not license restating a metastatic result as a resectable one. *Informs* means a simulated result shaped a design choice, and does not license presenting a simulated survival figure as an expected outcome. *Supports* means a verified computation raises confidence in a stated assumption, and does not substitute for a clinical measurement. *Specifies* means a bench measurement fixes a numeric requirement, and does not claim the requirement has been met in a patient. *Bounds* means a result sets a monitoring limit, and does not make that limit a predicted rate.

6.4 What the Work Cost, and What It Costs Elsewhere

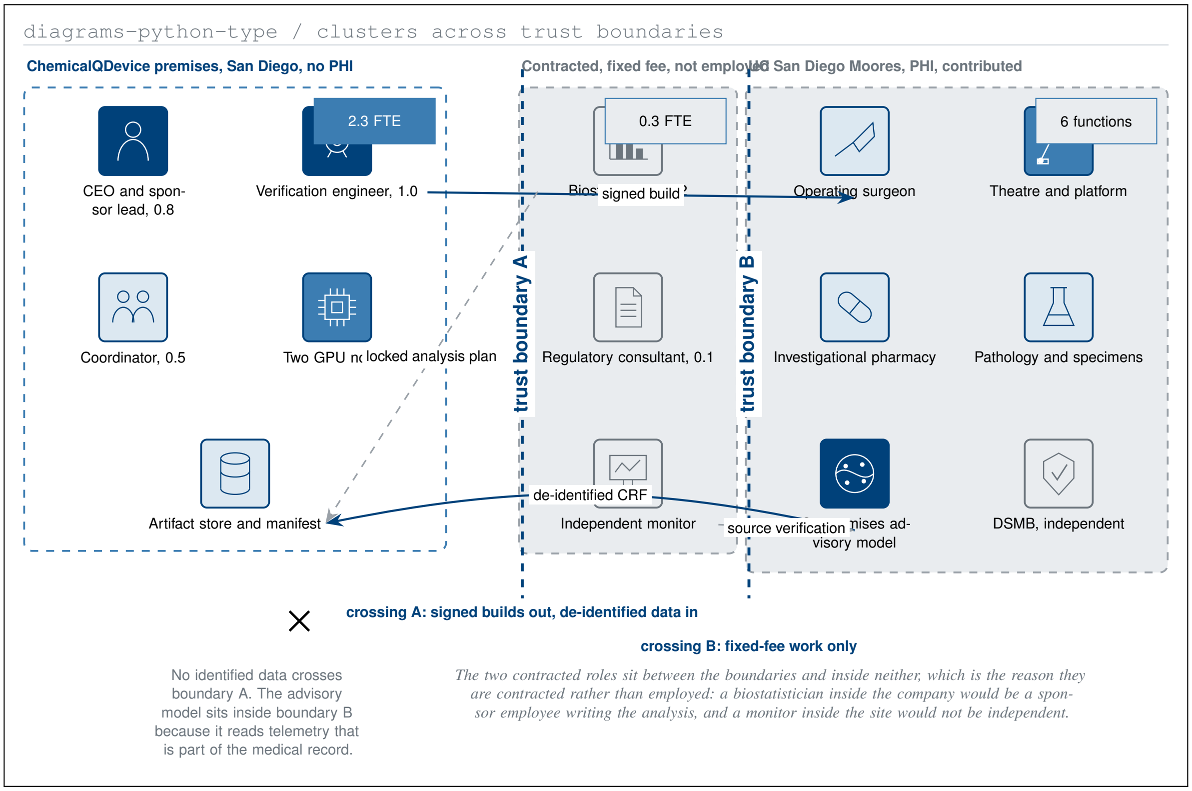
Work product	Industry cost	Industry schedule	This author
Empirical virtual trial, triplicate	above $120,000 per run	4.5 months	$36,330 estimated, about one month
Ten-arm QSP virtual trial	above $2,000,000	several months to over a year	$36,330 estimated, about one month
Digital-twin simulator platform	$28,000 to $700,000	3 to 16 weeks	$36,330 estimated, about one month

Three work products against their industry cost and schedule benchmarks. The author figures are estimates from the source set and are reported as such, not as audited accounts.



Five published quantities and the exact filing section each one reaches. Every edge carries its link type, and every chain ends at a milestone whose artifact a program officer can request.

The same cost discipline that produced a $36,330 virtual trial produces a 7.5 percent indirect rate. Both are the same fact seen twice: a firm with no facilities, no administrative layer and no negotiated rate agreement has very little between an appropriation and the work it buys. Figure 3 prices that fact at $531,000 against a university route.



Three clusters and two trust boundaries. The company employs 2.3 of the 2.6 FTE, contracts 0.3, and pays for none of the six site functions. No identified data crosses the left trust boundary.

7 Small-Business Operating Plan

7.1 Who Does the Work

At full Phase II staffing the company runs on 2.6 full-time equivalents, of which 2.3 are employed and 0.3 are contracted at a fixed fee. At Phase I staffing the figure is 1.1: the chief executive at 0.6 and a verification engineer at 0.5. The clinical research coordinator is not hired until a participant can be consented, which is month 10.

Role	FTE	Status	Budget line	Boundary
CEO and sponsor lead	0.8	Employed	Phase II personnel	Company
Verification engineer	1.0	Employed	Phase II personnel	Company
Clinical research coordinator	0.5	Employed, from month 10	Phase II personnel	Company
Biostatistician	0.2	Contracted, fixed fee	Phase II consultants	Neither
Regulatory consultant	0.1	Contracted, fixed fee	Phase II consultants	Neither
Total	2.6	2.3 employed, 0.3 contracted	$600,000 personnel and $34,000 consultants	

The 2.6 FTE by employment status, the budget line that pays each, and the trust boundary each sits inside. The two contracted roles sit inside neither boundary, which is the point of contracting them.

7.2 Where the Work Happens

Six functions sit at the site and none is on the company payroll. The operating surgeon and assistant, the investigational pharmacy, and pathology with specimen handling are paid out of the site clinical conduct and laboratory lines against an executed clinical trial agreement. Theatre and robotic platform time and the host for the advisory model are contributed and unpriced. The data and safety monitoring board is convened by the site and paid from the independent monitoring line.

The advisory model runs on the hospital's side of the boundary rather than the company's, and that is an architectural choice rather than a convenience. The model reads intraoperative telemetry that forms part of the medical record. Moving the model to the company's premises would move identified data with it, and the company would then hold protected health information that it has no reason to hold.

7.3 How a One-Person Company Runs a Trial

The reviewer's real question is whether 2.6 people can run a first-in-human surgical trial. The answer is structural rather than rhetorical, and it has two halves.

The first half is the division of labour drawn in Figure 6. The company is the sponsor: it holds the IND, authors and amends the protocol, builds and verifies the interlock rig, files expedited safety reports, and deposits the archive. The site holds every clinical function and, more importantly, every clinical decision. The two are joined by an executed clinical trial agreement, which is the first row of the absent register in Table 7 and does not yet exist.

The second half is that the three most expensive things in a trial of this kind do not scale with headcount and are already finished. The protocol is written. The simulation stack is built. The verification suite is frozen and hashed. A company that had to produce those three inside the award would need a team; a company that produced them before applying does not.

7.4 Governance Inside the Company

Six arrangements govern a firm with one officer, and they are deliberately unglamorous. The chief executive is the sole officer and sole director. The data and safety monitoring board is independent and is convened by the site. The biostatistician is contracted at a fixed fee with the analysis plan locked before the first dose. An independent monitor is retained per visit. Annual financial statements are prepared by an outside accountant, which together with insurance is what the 7.5 percent indirect rate pays for. Records are retained under the sponsor and investigator terms of 21 CFR §312.57 and §312.62 [35].

One thing is deliberately absent. No scientific advisory board is proposed. A one-product company at 2.6 full-time equivalents cannot service an advisory board honestly inside thirty-three months, and a board that meets twice and signs nothing is decoration rather than governance.

8 San Diego and the August 2026 Record

8.1 Four Contacts, Dated

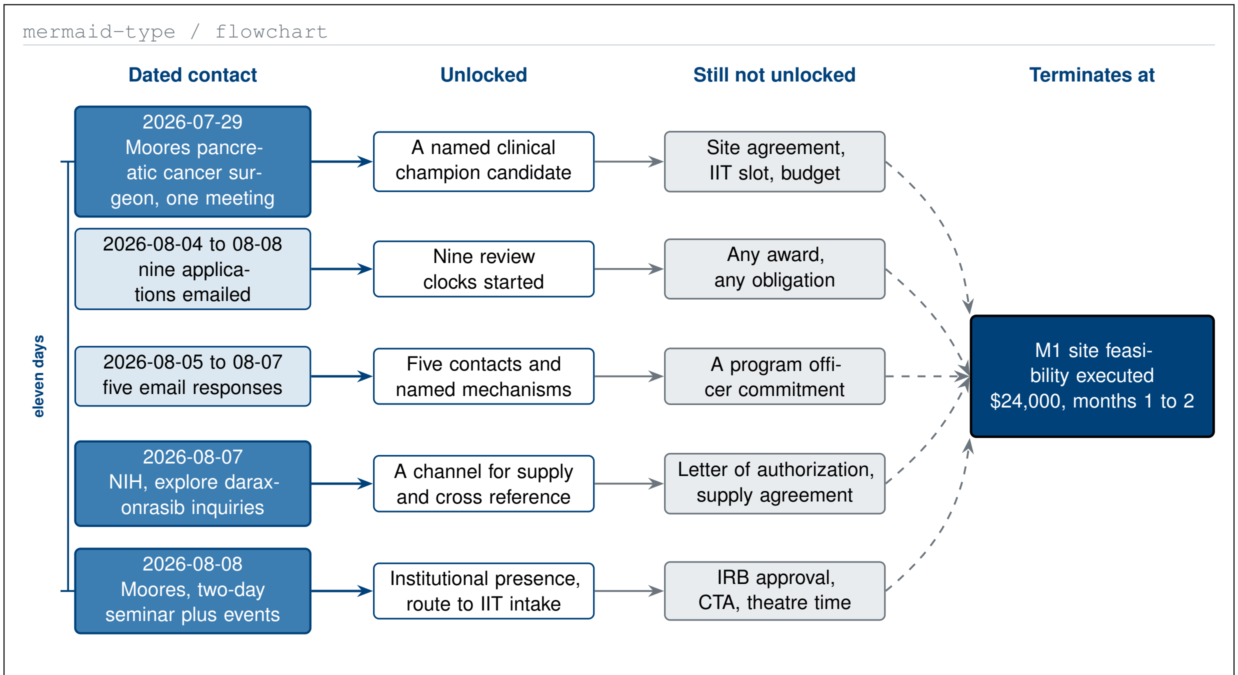
Every other section of this document describes a plan. This one describes a record, and the register changes accordingly: what follows is four contacts and five responses in eleven days, stated at their actual size.

On **29 July 2026** the author met one pancreatic cancer surgeon at UC San Diego Moores Cancer Center. One meeting, one surgeon.

Between **4 and 8 August 2026** nine of the ten funding applications were emailed to their recipients, with their attachments and cover text as deposited [1].

Between **5 and 7 August 2026** five email responses were received from those applications.

On **7 August 2026** an NIH email indicated willingness to explore additional pancreatic cancer inquiries concerning daraxonrasib.



Rows 2 and 3 are drawn paler than the other three because they record correspondence rather than a meeting, and the distinction has to survive without reading the labels. All five arrows terminate at the same milestone and not one of the five closes it.

Four contacts between July 29 and August 8, 2026, and what each one actually unlocks. Every arrow ends at the same milestone, M1, and not one of the four is on its own sufficient to close that.

On **8 August 2026** UC San Diego Moores Cancer Center emailed an offer of a two-day seminar together with associated seminar events.

8.2 What Each One Unlocks, and What It Does Not

The surgeon meeting unlocks a named clinical champion candidate. It does not unlock a site agreement, a slot in the investigator-initiated trial intake, or a budget.

The nine emailed applications start nine review clocks. They do not create an award or an obligation of any kind on any recipient.

The five responses identify five points of contact and five named mechanisms. They are not a program officer's commitment, and none of the five is a statement about this trial's merits.

The NIH daraxonrasib email opens a channel for inquiries about supply and regulatory cross-reference. It does not produce a letter of authorization or a drug supply agreement.

The Moores seminar offer opens institutional presence and a route toward the intake committee. It does not produce IRB approval, a clinical trial agreement, or theatre time.

8.3 Why the Institution Is the Constraint, Not the Money

The finding is carried over from the ten-application analysis rather than re-derived: the programme's schedule is governed by the institution and not by any funder, and an award arriving in month four cannot begin work until the site agreement executes [1]. That is guard G4 in Figure 7, the only guard the company cannot satisfy by working harder, and it is halt point H4 in Figure 14, the earliest of the five at month six.

It is also why the cheapest row in the entire milestone schedule is the first one. M1 costs $24,000, which is 1.5 percent of the award, and everything after it is downstream.

8.4 The NIH Daraxonrasib Channel

The 7 August email establishes a channel for inquiries about the agent. It is worth stating precisely what does not follow from it, because a reader scanning for traction will be tempted to read more into it than it contains. There is no letter of authorization. There is no drug supply agreement. There is no regulatory cross-reference to any existing IND. There has been no communication of any kind with the agent's developer.

Two rows of the absent register in Table 7 would eventually be closed through this channel: the drug supply agreement, which blocks M6, and the letter of authorization, which blocks M5. Revolution Medicines is the only signatory for both, and no approach has been made.

8.5 The Seminar, and What It Is For

The 8 August offer of a two-day seminar with associated events is the most useful of the four contacts, for a reason that has nothing to do with the science. It is the route to institutional presence and, through it, to the investigator-initiated trial intake committee, which is the body that controls M1.

All four contacts advance M1. None closes it. M1 costs $24,000 and is the cheapest row in Table 14, and until it closes nothing else in this document can begin.

9 Risks, Stop Conditions, and What This Is Not

9.1 What the Conversion Costs

Four things are worse about this document than about the ten it replaces. Each is stated first and answered second, in that order.

1. The individual-scientist framing is abandoned. It is what made all ten prior applications distinctive, and it is the framing the report rewards most explicitly. The answer is that the framing was never the subject; the trial was. An entity applying under a person-based mechanism would be applying under the wrong clause, and §1 shows the eligibility test it fails.

2. Capitalization detail is published that the author might prefer to keep private. Table 13 states the founder's holding at each of five stages, Table 9 and Table 10 state every line of both budgets, and all of it is permanent under a DOI. The answer is that a milestone schedule with a redacted budget is not auditable, and an unauditable schedule is the thing this document exists to replace.

3. Three of the ten prior recipients will read this as off-mechanism. The NIH Director's Pioneer Award and the HHMI Investigator Program are person-based, and the focused research organization route is organization-based and time-bound to dissolution. A company plan is the wrong document for all three. The answer is that it was always the wrong document for all three, and §1 is the argument that it is the right one for the fourth.

4. A financing paper is not a scientific publication and fits no journal. It has no hypothesis, no method, and no result in the sense a journal uses those words. The answer is that Zenodo is the correct venue for it and the only one claimed, and that the paper is labelled a plan on its cover.

9.2 The Risk Register

Risk	Attaches to	Likelihood	Mitigation, and whether it exists today
Site agreement does not execute	H4, G4, M1	High	None today. Requires an executed CTA that no company action produces
IND clinical hold issued	H1, G3, M5	Low	The IND package is drafted and deposited; the 30-day clock is the only unknown
Two of three DLTs at dose level 1	H2, M8	Moderate	Exists: 3+3 de-escalation is written into the protocol with a defined dose range
Phase II award is not made	H3, M6	Moderate	Exists: the Descoped state resubmits at three participants rather than six
Founder falls below 50 percent	H5, stage 4	Low inside 33 months	Exists: every tier 3 instrument is gated behind a milestone, and stage 4 sits outside the plan
Investigator acquires a §54.2 interest	H5	Low	Exists: the investigator role is assigned wholly to the site and each files Form 3454
Advisory output mistaken for a directive	M7	Moderate	Exists: the model is confined to advisory output and every arrow reaching the robot leaves a human first
Bench p95 latency above 250 ms	H1, G1, M3	Low	Exists: the interlock rig is a Phase I deliverable measured before any participant
The 81.9 credibility score fails to reproduce	M4, WP15	Low	Partial: the suite is hashed at M4, but the full re-run is WP15 and is unfunded at $1,606,000
No alternate San Diego site accepts	H4	Moderate	None today. The plan names one site and holds no fallback agreement

Ten risks, the halt point or milestone each attaches to, and whether the mitigation exists today or is merely planned. Three rows read none today or partial, and those three are the honest content of the table.

Three rows carry no mitigation that exists today. Two of the three are the same fact seen twice: the plan depends on one institution and holds no alternative. The third, the verification re-run, is unfunded because it sits in the $2,104,000 the SBIR route does not reach, which is the kind of consequence §3 exists to make visible rather than to bury.

9.3 What This Document Is Not

It is not a submission of record. It is not an agreement with UC San Diego, Moores Cancer Center, or any other institution, and no agreement of any kind exists. It is not a completed raise: no term sheet, simple agreement for future equity, or subscription agreement exists, and every capitalization figure in Table 13 is a plan. It is not a claim that daraxonrasib is first in human; the agent is investigational and already in Phase 3 evaluation. It is not medical or regulatory advice. No robotic configuration has been specified or cleared. No patient has been treated.

9.4 The Two Single Points of Failure

Figure 14 makes them visible, and a reader who has skipped the figure should still meet them here.

H1, the month-nine gate, is an OR branch. It fails on either a bench stop latency above 250 ms at the 95th percentile or an IND clinical hold. Either alone ends Phase I without a Phase II request.

H5, the ownership test, is the other OR branch. It fails on either the founder falling below 50 percent or a clinical investigator acquiring an interest under 21 CFR §54.2. Either alone breaks the eligibility the whole plan rests on.

Neither is technical in the sense that more engineering fixes it. One is regulatory and instrumental; the other is a fact about who owns the shares. Both are knowable in advance and both are measured before they can do damage, which is the only reason a schedule carrying two single points of failure is worth putting in front of a program officer.

10 Build Method and Reproducibility

10.1 The Eight-Stage Build

The paper was built as one schedule of eight stages: five diagram-specification stages, then a draft, a full and a final paper stage. Two rules shaped it. One commit per file at the moment the file is written, so a reader can watch the branch rather than wait for it. And no Python file anywhere, so the repository's three lint-and-format jobs stay green; the diagrams stage emits a machine-readable specification instead, and every tile it describes is drawn in LaTeX by a vector glyph macro.

Stage	Output directory	Figures or sections produced	Commits
1	`mermaid/`	Figures 1, 7, 12, 13, 19	6
2	`plantuml/`	Figures 6, 10, 15	4
3	`d2/`	Figures 2, 5, 8, 11, 16	6
4	`diagrams-python/`	Figures 4, 18, 20	4
5	`graphviz/`	Figures 3, 9, 14, 17	5
6	`draft-capital/`	Twelve sections, twenty sized slots	10 or more
7	`full-capital/`	Twenty figures drawn, twenty-one tables	10 or more
8	`final-capital/`	The senior-author pass	10 or more

The eight stages, their output directory and their commit count. There is no publication subdirectory at stage 8, by instruction, and no stage batches its work to the end.

10.2 Why the Diagram Split Is Uneven

Five mermaid, three plantuml, five d2, three diagrams, four graphviz. The split follows purpose rather than quota, and the two five-counts fall where this paper argues most often. Mermaid answers what happens next and how long, and a capitalization plan is mostly a claim about sequence. D2 answers what contains what and what is priced against what, and the rest of the plan is tables. The per-figure reasoning is in the five directory READMEs and is not restated here.

10.3 What Survives a Stop

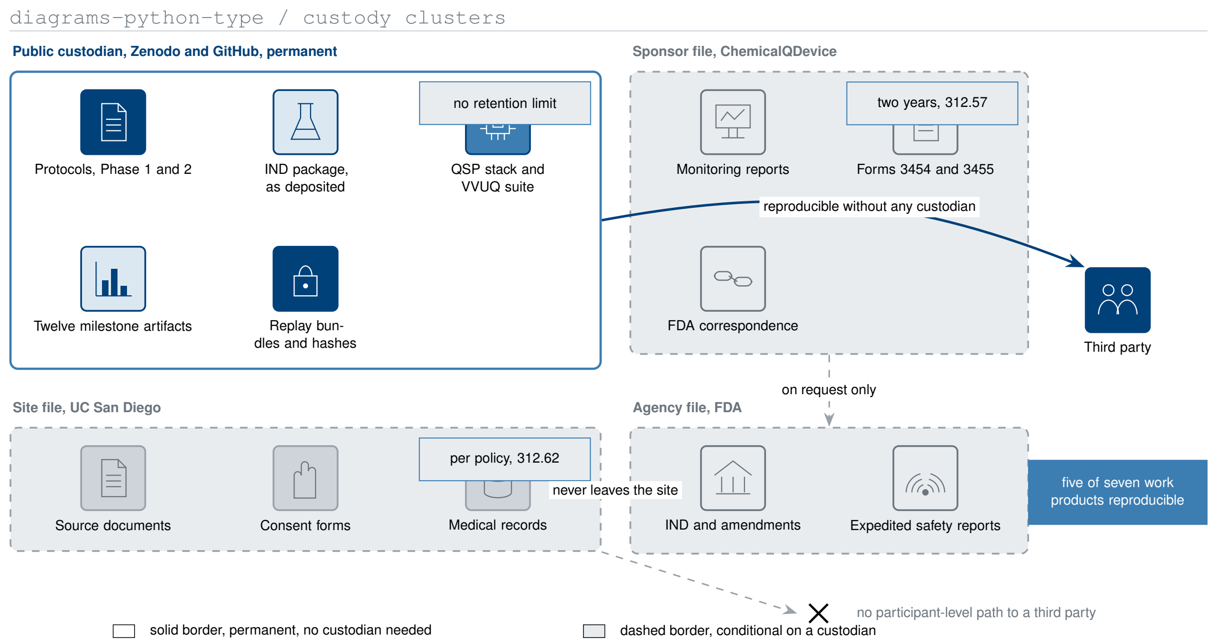
Five of seven work products are fully reproducible from the public custodian alone: the ten-arm simulation, the 1000-patient digital twin, the 55-test verification suite with its credibility score, the replay of any advisory decision, and this paper with all twenty figures. The interlock bench verification is partly reproducible, since the report and the manifest are public but the rig is not. Only participant-level clinical results are unreachable, because the source documents never leave the site [35].

That is the argument this figure exists to make, and it is made once. If the company dissolves at any milestone, the computational work does not become unavailable, because it was never in the company's custody in the first place [41].

10.4 Verification of This Document

Every source set in this build was compiled with pdfLaTeX twice and BibTeX before its commit, at zero errors, zero overfull boxes, zero undefined citations, and zero undefined references. No raster image is used anywhere in this paper: every figure is pure TikZ on an eight-token palette with no black fill, and the palette audit is a search for the deleted near-black fill token, which returns nothing.

The figure-to-caption distance is identical at 6.06 pt for all twenty figures and all twenty-one tables. The audit that proves it is arithmetic rather than visual: the count of `\vspace{-0.65cm}` in the section sources equals the count of `\figcaption` plus the count of `\tabcap`, and the style file closes both carriers with a single rigid skip that no section can override.



The public cluster's border is solid and the other three are dashed: solid is permanent, dashed is conditional on somebody continuing to hold the file. The blocked edge is drawn to a third of its length and terminated, so it reads as a path that does not complete.

Four custodians and what each still holds if the programme stops. Everything a third party needs to reproduce the computation sits with the first custodian, which needs no one to stay in business.

Nine defects were found and fixed during the build rather than carried forward, and they are recorded because a verification section that reports only successes is not a record. Four were measurement failures in the source: a three-cell cover ledger 14.34 pt past the measure, four table columns whose bold headers were wider than any body cell in them, eleven digital object identifiers overflowing a 4.0 cm column because the printed form is unbreakable, and a directory path in a monospace face overflowing by 11.23 pt because Courier offers no break point inside a path. Two were syntax: a TikZ node width written as `{\w*0.415} cm` is parsed by the mathematics engine as one expression and fails, the unit belonging inside the braces, and a `\foreach` loop iterated a list with an empty body. Two were layout: a 143 pt overfull vertical box where a figure, a table and a closing paragraph competed for one page, and a float budget inherited from the parent work that allowed three floats on a page this paper's figures are too tall for. The last was arithmetic: the draft index counted nineteen tables where the paper carries twenty-one, and sixteen cross-references were renumbered against the corrected index.

Back Matter

Abbreviations

Short	Expansion	Short	Expansion
AE	Adverse event	LLM	Large language model
CRC	Clinical research coordinator	LOA	Letter of authorization
CRF	Case report form	MTD	Maximum tolerated dose
CTA	Clinical trial agreement	MTDC	Modified total direct cost
CTCAE	Common Terminology Criteria for Adverse Events	NIH	National Institutes of Health
ctDNA	Circulating tumour DNA	PDAC	Pancreatic ductal adenocarcinoma
DLT	Dose-limiting toxicity	PD	Pharmacodynamics
DOI	Digital object identifier	PK	Pharmacokinetics
DSMB	Data and safety monitoring board	QSP	Quantitative systems pharmacology
F and A	Facilities and administrative cost	RP2D	Recommended Phase 2 dose
FDA	Food and Drug Administration	SAE	Serious adverse event
FRO	Focused research organization	SAFE	Simple agreement for future equity
FTE	Full-time equivalent	SBA	Small Business Administration
IIT	Investigator-initiated trial	SBIR	Small Business Innovation Research
IND	Investigational new drug	STTR	Small Business Technology Transfer
IRB	Institutional review board	VCOC	Venture capital operating company
ISGPS	International Study Group of Pancreatic Surgery	VVUQ	Verification, validation and uncertainty quantification

Abbreviations used in this paper, set in two column pairs at the body measure. Trial-specific terms follow the definitions in the published Phase 1 protocol rather than being redefined here.

Data and Code Availability

Every work cited to a digital object identifier is openly deposited. The repository is at `https://github.com/kevinkawchak/physical-ai-oncology-trials`, version v4.5.0 [41].

The directory `funding/capitalization-plan` contains the eight-stage sub-prompt schedule, the twenty figure specifications across five machine-readable diagram platforms, the three build stages with an Overleaf bundle for each, the master prompt filed verbatim with the full build output, and the TikZ source of every figure printed here. No raster image is used anywhere in this paper.

Author Contributions and Conflicts

Kevin Kawchak is the sole author. He is the sole officer, sole director and sole owner of ChemicalQDevice, which is the sponsor described throughout this document, and he holds no equity in Revolution Medicines or in any robotic surgery vendor. He is not, and under this plan will not become, a clinical investigator on the trial: every trigger in 21 CFR §54.2 that his position would fire is the reason §4 of this paper exists, and the investigator role is assigned wholly to the site. No term sheet, simple agreement for future equity, or subscription agreement exists, and every capitalization figure in Table 13 is a plan rather than a completed transaction. No sponsor, contract research organization, institutional review board, regulator, or medical society reviewed this paper or authorised any statement in it. The work was adapted using Claude Code Opus 5.

Citation

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